Debre Berhan University



College of Computing

Department of Software Engineering

Internship Report

Project Title: Mirchaye Election Platform

Hosting Company: Ewenet Communication PLC

By: Biruk Ashagre 1401048

Abdela Nesredin 1400751

Mentor: Ayana F. and Alebachew Ch.

Company Supervisor: Samuel Abera

Sept, 2025

Acknowledgment

First and foremost, we'd like to express our heartfelt gratitude to all computing department staff and our mentors for being an incredible helper to our internship. Your unwavering support and guidance mean the world to us we also want to give a big shout-out to our parents. Your endless encouragement and support have been our rock throughout this internship.

Of course, we can't forget the fantastic team at the Ewenet Communication PLC. You all have been amazing mentors, sharing your wisdom and making us part of your awesome team. We're so thankful for the opportunities you've given us.

We're super grateful for the trust and confidence you've had in us. We promise to carry these invaluable lessons and experiences with us throughout our career. Your support has been instrumental in our growth, and for that, we're deeply appreciative.

With heaps of gratitude,

Definition, acronyms, abbreviation

NEBE: National Election Board of Ethiopia

CRUD:Create, Read, Update, Delete

RBAC: Role-Based Access Control

ERP: Enterprise Resource Planning

API: Application Programming Interface

CRM: Customer Relationship Management

UI/UX: User Interface/User Experience

MVC: Model-View-Controller

Document organization

Part one: Internship experience

Part Two: Real world Project (including SRS and System Design).

Contents

Executive Summary 6
INTRODUCTION 7
1. Introduction 7
1.1 Background of the company 8
1.2 Mission of the company 8
1.3 Vision of the company 9
1.4 Goal of the company 9
1.5 Main product and service of the company 10
1.6 Main customers and end users of company 10
2 Overall Internship Description and Activities 11
2.1 Objective of the internship 11
2.1.1 General Objective 11
2.1.2 Specific Objective 11
2.2 Nature of work/ task carried out 12
2.2.1 Week one activities 12
2.2.2 Week two activities 12
2.2.3 Week three activities 13
2.2.4 Week four activities 13
2.2.5 Week five activities 13
2.2.6 Week six activities 14
2.2.7 Week seven and eight activities 14
2.2.8 Week nine and ten activities 15
2.2.9 Week eleven and twelve activities 16
2.2.10 Week thirteen and fourteen activities 16
2.3 Section we have been working on 16
3. Experience and transferrable skill 18
3.1 Work experience 18
3.1.1 Technical experience 19
3.1.2 Non-technical experience 21
3.2 Benefits of experience 22
3.3 Skills you get from experience (soft and hard skills) 23
3.4 Challenges you face and Solution you have taken (personal and company challenges) 25
4 General Overview of the project 28
4.1 Background of the project 28

4.2 Objective of the project 29
4.2.1 General objective 29
4.2.2 Specific objective 29
4.3 Statement of the problem for the project..... 30
4.4 Methodology 32
4.5 Development tool 35
4.6 Functional Requirements: 39
4.6.1 Non-Functional Requirements: 41
4.7 Diagram..... 42
4.7.1 Usecase diagram 42
4.7.2 Sequence Diagram 43
4.7.3 Class Diagram 48
4.7.4 Component Diagram 49
4.7.5 Deployment Diagram 50
4.7.6 Database Design 51
5 Conclusion and recommendation 52
5.1 Conclusion 52
5.2 Recommendation for Future Internship Program: 53
5.2.1 Recommendation for Future Internship Program: 53
5.2.2 Recommendation for Hosting Company:..... 54
5.2.3 Recommendation for Department: 54
5.2.4 Recommendation for University: 55
6 Appendix 56
7. Reference 59

List of Tables

Table 1: Scope of the document..... 33
Table 2: Tool used by the project..... 35
Table 3: Non-Functional requirement..... 41

List of Figures

Figure 1: work flow of Existing System..... 37

Executive Summary

The Mirchaye is a digital platform developed to address a critical need during national elections in Ethiopia: access to trusted, real-time, and multilingual election information. It is built with a strong commitment to user privacy, political neutrality, and digital accessibility. The platform serves the general public, political parties, and NEBE officials, offering features like election news, political party updates, profiles, and educational content — all without requiring users to register or submit personal information. Ultimately, it aims to foster transparency, fight misinformation, and strengthen voter engagement across Ethiopia.

Mirchaye is more than just an information platform—it is a tool to combat misinformation, foster civic education, and strengthen democratic engagement. By providing transparent and inclusive access to electoral content, it aims to build a more informed and empowered electorate.

As Ethiopia continues its democratic journey, the Mirchaye platform positions itself as a vital civic resource—advancing transparency, public trust, and equitable participation in national decision-making.

INTRODUCTION

1. Introduction

Elections are fundamental democratic processes that require transparency, accessibility, and public trust to function effectively. In many countries, including Ethiopia, electoral environments often face challenges such as fragmented information, limited access to credible sources, and the spread of misinformation—all of which can undermine the integrity of elections and voter confidence.

Ethiopia, electoral information is frequently dispersed across multiple channels, making it difficult for citizens to distinguish between verified updates and political propaganda. Disparities in access to information—due to geographical, linguistic, and technological barriers—further exacerbate these challenges. Many citizens, particularly those in rural or underserved communities, lack direct and consistent access to official election updates, civic education, or neutral political party information.

To address these issues, the Mirchaye platform was developed as a centralized, secure, and impartial digital platform designed to streamline electoral communication and enhance public engagement. Unlike traditional media or social platforms, Mirchaye operates as a non-partisan tool, ensuring that all users—regardless of location or language—have equal access to verified election information without requiring registration or compromising personal data.

Key Features of the Mirchaye App:

- For Political Parties and Individual**:** A dedicated space to share campaign messages, manifestos, and official updates, ensuring fair and direct engagement with voters.
- For Citizens: Easy access to verified party profiles, real-time election updates, and NEBE announcements in multiple local languages, fostering informed decision-making.
- For the National Election Board of Ethiopia (NEBE): A direct channel to disseminate official guidelines ,news and critical announcements—bypassing media distortions or third-party interference.

By bridging the gap between voters, political stakeholders, and electoral authorities, the Mirchaye platform reduces misinformation, enhances transparency, and strengthens democratic participation. It is not merely a technological solution but a civic tool designed to elevate electoral integrity and public trust in Ethiopia's democratic processes.

Ultimately, the Mirchaye platform represents a proactive innovation in electoral communication, ensuring that every citizen—regardless of language, location, or digital literacy—can participate in elections with confidence, knowledge and clarity

1.1 Background of the company

Ewenet Communication PLC is a pioneering technology and digital solutions company headquartered in Addis Ababa, Ethiopia. Established in 2012 by three visionary young entrepreneurs, Ewenet has rapidly evolved from a modest startup with 20,000 ETB capital into a leading innovator in software development, branding, and digital transformation across Africa. The company's multifaceted expertise underscores its pivotal role in Ethiopia's tech ecosystem.

Operating under Ethiopian business law, Ewenet specializes in software development, mobile applications, web design, social media marketing, and multimedia production. Its organizational framework comprises dedicated teams for creative design, programming, and digital strategy, ensuring comprehensive solutions for clients.

Beyond commercial services, Ewenet actively contributes to civic tech initiatives, such as developing the Mirchaye election platform to promote transparent political campaigns. The company's commitment to innovation is further demonstrated through its COVID-19 response tools and the Ewenet *Stage* talent-empowerment platform.

Embracing cutting-edge technology, Ewenet integrates modern tools like AI-driven design software and secure cloud-based development to deliver efficient, scalable solutions. Professional growth is a core focus, with regular training programs ensuring its team remains at the forefront of industry trends.

1.2 Mission of the company

The mission of Ewenet Communication PLC is to leverage technology and creativity to build a digitally empowered Africa. This mission is encapsulated as follows:

"To deliver innovative, user-centric digital solutions that exceed client expectations while fostering societal progress. Ewenet is dedicated to advancing Ethiopia’s tech landscape through software development, branding, and civic tech initiatives. By combining technical excellence with creative vision, we strive to bridge gaps in accessibility, education, and opportunity—empowering businesses, governments, and individuals to thrive in a digital era."

In essence, Ewenet’s mission aligns with its role as a catalyst for digital transformation, youth empowerment, and inclusive growth in Ethiopia and beyond.

1.3 Vision of the company

Looking ahead to 2024 E.C, Ewenet Communication PLC envisions becoming the premier technology and digital solutions provider in East Africa, recognized for its innovation, reliability, and social impact. We aim to establish ourselves as the go-to partner for businesses, government institutions, and international organizations seeking cutting-edge digital solutions. Our vision includes creating a robust ecosystem that nurtures African tech talent through our Ewenet Stage platform and internship programs. We strive to be at the forefront of Ethiopia's digital revolution, developing solutions that address critical challenges in governance, education, and commerce while maintaining the highest standards of quality and user experience.

1.4 Goal of the company

Ewenet Communication PLC operates with three primary objectives that guide our operations and growth. First, we aim to develop innovative, scalable technology solutions that address pressing societal needs, as evidenced by our work on election systems and civic health tools. Second, we are committed to delivering superior digital products and services that help businesses and organizations establish and enhance their digital presence. Third, we focus on creating meaningful impact through youth empowerment initiatives, including our internship programs and collaborations with educational institutions. These goals collectively support Ethiopia's economic development, digital literacy improvement, and the creation of a vibrant tech ecosystem.

1.5 Main product and service of the company

Ewenet Communication PLC offers a comprehensive range of digital products and services tailored to meet the evolving needs of modern businesses and institutions. Our software development services include custom ERP and CRM systems, as well as specialized in developing platforms . We provide complete digital branding solutions encompassing logo design, corporate identity development, and social media strategy. Our web and mobile application development services help clients establish robust online presences, while our multimedia division delivers high-quality video production and animation services. Additionally, we conduct technology training programs designed to equip students and professionals with in-demand digital skills.

1.6 Main customers and end users of compan

Ewenet Communication PLC serves a diverse clientele that spans multiple sectors and user groups. Our primary customers include private businesses seeking to establish or enhance their digital presence, NGOs requiring specialized technological solutions, and government agencies like the Ministry of Labor, Ministry of Health and Social Affairs that we've assisted with rebranding initiatives. The end users of our products range from Ethiopian citizens accessing our civic technology platforms to African professionals utilizing our talent development services through Ewenet Stage. We also collaborate with educational institutions and startups, providing them with the technological tools and expertise needed to thrive in the digital economy. This broad spectrum of clients and users reflects our commitment to serving Ethiopia's growing digital needs while contributing to the continent's technological advancement.

2 Overall Internship Description and Activities

2.1 Objective of the internship

2.1.1 General Objective

The primary aim of the internship was to gain hands-on experience in developing, managing, and optimizing the Mirchaye platform, a digital platform designed to enhance electoral transparency and civic engagement in Ethiopia. This internship provided an opportunity to contribute to a technologically innovative solution that addresses critical challenges in Ethiopia's electoral process, including misinformation, limited access to credible sources, and voter education. The overarching goal was to understand how digital tools can strengthen democratic participation while ensuring security, accessibility, and neutrality in electoral communication.

2.1.2 Specific Objective

- Platform Functionality and Development: To comprehend the technical architecture of the Mirchaye platform, including its backend systems, user interface design, and data integration processes that enable real-time election updates from official sources like the National Election Board of Ethiopia (NEBE) and political parties and individual candidates who are representatives in House of people representative.
- Legal and Regulatory Compliance: To familiarize with Ethiopia's electoral laws and digital communication policies, ensuring the platform adheres to guidelines for neutrality, data protection, and content moderation.
- User-Centric Design and Accessibility: To analyze how the platform caters to diverse users, including rural populations and first-time voters, through multilingual support, offline functionality, and intuitive navigation.
- Content Verification and Anti-Disengagement Measures: To assist in developing protocols for verifying and publishing content from political parties and NEBE, ensuring all information is accurate, unbiased, and free from manipulation.

- Privacy and Security Safeguards: To study the platform privacy-by-design approach, including its no-login requirement, which protect users from surveillance or political targeting.

- Stakeholder Engagement and Training: To support training programs for political parties and NEBE officials on using the platform to disseminate updates, ensuring effective and equitable communication with voters.

- Impact Assessment and Recommendations: To evaluate user engagement metrics and feedback, identifying opportunities to enhance the platform reach, usability, and impact on voter education.

2.2 Nature of work/ task carried out

2.2.1 Week one activities

During the first week at Ewenet Communication, we familiarized ourselves with the company's work environment, rules, and the core technologies in use. We were introduced to Laravel development and assigned a project where our primary focus was on backend engineering. Our task involved developing APIs for a test website, which required us to learn the Laravel framework, design database schemas, and engage in collaborative problem-solving. We attended orientation sessions that outlined the internship schedule and regulations, confirming our working hours from 2:30 PM to 11:30 PM, Monday through Friday. We also collaborated with our mentor and team to understand project expectations and manage Git workflows. Outside of company hours, we dedicated time to practicing the Blade templating engine for server-side rendering and reviewed tutorials to prepare for the upcoming week's tasks.

2.2.2 Week two activities

This week deepened our Laravel expertise, particularly in authentication (Laravel Breeze), authorization (Gates/Policies), and database integration (SQLite, MySQL, MongoDB). we configured MongoDB for testing and explored Laravel-MongoDB packages. Collaborating with a peer, we initiated a News API project, designing its schema and repository. Over the weekend, we implemented role-based middleware and studied Laravel Sanctum for API authentication, aligning these skills with the company's backend requirements.

2.2.3 Week three activities

In the third week, we made significant progress on a full-stack Laravel news portal project. We developed robust backend CRUD operations, implemented multi-user authentication with role-based access control (admin, editors, reporters), and optimized database performance using Eloquent relationships and caching. On the frontend, we built dynamic, responsive views using the Blade templating engine, integrated server-side pagination, and created a secure file upload system for article thumbnails. We also constructed an admin dashboard and implemented features like search functionality and audit logging. Outside of work, we experimented with advanced features like Laravel Livewire, a recommendation engine, and SEO implementation to enhance the project's functionality.

2.2.4 Week four activities

A pivotal week, as we successfully launched and presented the news portal to my mentor, showcasing backend APIs and frontend integration. Subsequently, we were assigned to the Mirchaye Election Information System for Ethiopia’s National Election Board (NEBE). We drafted the Software Requirements Specification (SRS) and Business Model Canvas, outlining features like multilingual content, user roles (NEBE/parties/citizens), and technical stacks (Flutter/Laravel/React(Next js)). Feedback from stakeholders guided refinements to the SRS, and we began designing the database schema and Flutter UI wireframes.

2.2.5 Week five activities

During the fifth week, we focused exclusively on core backend development for the Mirchaye project. We built the Laravel backend to support multiple user roles (NEBE, political parties, Individuals) and implemented multilingual content management for posts, news, and announcements. Key accomplishments included developing secure API endpoints using Laravel Passport for authentication, implementing robust CRUD operations for party and candidate data, and integrating localization logic to serve content in different languages. We conducted thorough unit testing and collaborated with the frontend team to ensure API responses met UI requirements, laying a strong foundation for the mobile and web applications.

2.2.6 Week six activities

The sixth week was dedicated to refining the backend API based on mentor feedback and beginning our exploration of the Next.js framework for the upcoming admin panels. We reviewed and discussed improvements to our API's role separation, validation, and route organization. Concurrently, we started studying core Next.js concepts such as file-based routing, layout components, and responsive image handling with next/image. This foundational work prepared us to plan the UI structure for the NEBE and political party admin dashboards, ensuring our backend refinements are perfectly aligned with the needs of the future frontend interfaces.

2.2.7 Week seven and eight activities

During this pivotal and intensely focused development cycle, our team's concerted efforts were channeled exclusively into the meticulous construction, refinement, and fortification of the core Laravel backend API, which has now been successfully elevated to a production-ready state, thereby establishing a robust and high-performance foundation for the entire Mirchaye ecosystem. This critical undertaking involved extensive architectural optimization where we meticulously profiled and tuned every major query, culminating in the strategic implementation of sophisticated data caching mechanisms for high-frequency endpoints—most notably for candidate profiles, political party information, and preliminary electoral data—a decisive intervention that has yielded a substantial and empirically verified reduction in direct database load, thereby dramatically enhancing the system's ability to gracefully handle the immense traffic surges anticipated during the election period. Concurrently, we undertook a comprehensive security overhaul, moving beyond basic authentication to institute a multi-layered defense strategy; this included the deployment of granular, IP-based rate limiting on all sensitive endpoints to ruthlessly mitigate brute-force and DDoS attempts, the hardening of all data

sanctification pipelines to prevent injection attacks, and the rigorous auditing of all authorization checks to ensure absolute data integrity and privacy for end-users, administrators, and political entities alike. With this powerful, scalable, and secure engine now complete and thoroughly validated, we have conclusively laid the essential groundwork upon which the Flutter mobile application for citizens and the Next.js citizen web portal will be immediately developed,

ensuring all future front-end development can proceed with confidence, leveraging a stable and performant data backbone.

2.2.8 Week nine and ten activities

Transition to staging environment marked a significant milestone in our project timeline. We configured a dedicated cPanel account mirroring our production specifications, where we first deployed the Laravel backend. This process involved meticulous attention to server requirements, including PHP version compatibility, necessary extensions, and proper file permissions. The database migration required special care to preserve sensitive voter information while maintaining referential integrity. For the frontend, we utilized cPanel's built-in tools to deploy the static Next.js export, addressing challenges with routing and asset loading through careful .htaccess configuration. Rigorous testing in this environment revealed several critical issues, particularly with file uploads for candidate documentation, which we resolved by adjusting PHP configuration limits and implementing chunked uploads for larger files.

2.2.9 Week eleven and twelve activities

The production deployment phase spanned these two weeks as we methodically transitioned from staging to live environment. We implemented a phased approach, beginning with the backend services on a dedicated subdomain (www.app.mirchaye.info) to isolate potential issues. The database migration required an overnight maintenance window to ensure data consistency, during which we implemented a read-only mode for the existing system. Frontend deployment followed, with special attention to cache management strategies to ensure users would receive timely updates during the election period. We conducted comprehensive smoke tests across all system components, verifying everything from user authentication to real-time result updates. The team developed and executed a detailed rollback plan, though fortunately it wasn't needed as the deployment proceeded smoothly.

2.2.10 Week thirteen and fourteen activities

With the system now live, our focus shifted to monitoring and performance tuning. We established daily checkpoints to review server resource utilization, identifying and addressing several memory leaks in our queue processing system. The cPanel monitoring tools proved invaluable during this phase, providing clear visibility into traffic patterns and resource usage. We implemented additional caching layers for geographic-specific content and optimized our database indexes based on actual usage patterns. One significant challenge emerged with timezone handling for election deadlines, which we resolved through coordinated updates to both frontend and backend time handling. We also conducted the first round of staff training sessions, creating detailed documentation for common administrative tasks and troubleshooting procedures.

2.2.11 Weeks 15-16: Project Handover and Documentation

The final weeks were dedicated to knowledge transfer and project closure. We compiled comprehensive system documentation, including architectural diagrams, API specifications, and operational procedures. The handover process included multiple training sessions, covering routine maintenance tasks, backup procedures, and basic troubleshooting. We created a series of video tutorials demonstrating common administrative workflows in the system. Performance benchmarks conducted at this stage showed the system comfortably handling concurrent users with sub-second response times for critical paths. The project concluded with a formal presentation to NEBE leadership, showcasing the system's capabilities and our development process. In retrospectives, we identified several key learnings, particularly around the importance of early environment parity between development and production setups, which would inform our approach to future projects.

2.3 Section we have been working on

During our internship , we contributed to all phases of the software development lifecycle from requirements analysis through deployment. This collaborative environment allowed us to learn industry best practices while applying our academic knowledge to a real-world project.

During the development of the Mirchaye Election Information Platform, our team employed multiple methodologies to ensure the system would effectively serve all stakeholders in Ethiopia's electoral process. The requirements gathering phase proved particularly crucial for shaping the platform's architecture and functionality.

For our interview process, we engaged in extensive discussions with key team members who brought diverse perspectives to the project. Samuel Abera, serving as our helper Backend Enginner and System Architect, provided critical insights into the technical constraints and security requirements necessary for handling sensitive election related informations. His expertise guided our API design decisions and database structuring. Tewolde Daniel Architect and Graphic Designer, our Design guider and UI/UX Specialist. These conversations were instrumental in designing distinct dashboard experiences for NEBE administrators, political party representatives and individual, and general citizens. Addis Alemayehu Founder & General Manager, acting as our Project Coordinator and NEBE Liaison, bridged the gap between technical implementation and operational realities, clarifying legal compliance requirements and communication protocols that would govern the platform's use during actual elections.

The document analysis phase revealed the intricate paperwork underlying Ethiopian electoral processes. We carefully examined election candidate registration forms, which informed our digital submission workflow design.

Our observational research uncovered several critical pain points in current election information management. This observation directly led to our implementation of real-time app and web updates.. Perhaps most importantly, we recognized the digital divide affecting rural voters, which prompted our optimization of the platform for low-bandwidth usage and offline caching capabilities.

These methodologies collectively shaped a platform that not only meets technical requirements but also addresses real-world challenges in Ethiopian election administration. The insights gained through interviews established our technical foundation, document analysis ensured regulatory compliance, and on-the-ground observations guaranteed practical usability across Ethiopia's diverse communities. This comprehensive approach resulted in a system that streamlines digital workflows, accelerates information dissemination, and improves accessibility for all electoral participants.

3. Experience and transferrable skill

3.1 Work experience

Interning Our internship at Ewnet Communication PLC working on the Mirchaye Election Information Platform proved to be an extremely valuable professional experience. As a team, we had the opportunity to develop a comprehensive digital platform designed to provide election-related news and information to citizens across Ethiopia.

During this internship, we gained hands-on experience developing both the backend administration system and public-facing mobile applications. Working closely with our mentor and the Ewenet development team, we contributed to all phases of the software development lifecycle from requirements analysis through deployment. This collaborative environment allowed us to learn industry best practices while applying our academic knowledge to a real-world project.

A key component of our work involved implementing Scrum methodology, with structured sprint planning sessions every Saturday. These regular check-ins helped us maintain focus, address challenges promptly, and continuously improve our development process. The Agile approach proved particularly valuable for managing the project's multiple components and stakeholders.

We developed a robust Laravel backend system that enabled NEBE administrators and political parties to publish and manage election news content. This included creating secure authentication systems with role-based permissions and building intuitive content management interfaces. The backend also served API endpoints to our cross-platform mobile application and website. which we designed to deliver election news to both iOS and Android users without requiring login credentials.

The mobile application development presented interesting challenges that enhanced our technical skills. We implemented features like offline content caching to ensure accessibility across Ethiopia's diverse regions and network conditions. Seeing the application function successfully on different devices and operating systems was particularly rewarding.

This internship significantly strengthened our abilities in full-stack development, collaborative coding, and Agile project management. The experience of working on a nationally significant platform that will be used in upcoming elections gave us valuable perspective on software development's real-world impact. We're grateful for the professional growth opportunity and the chance to contribute to Ethiopia's electoral information ecosystem.

3.1.1 Technical experience

The Mirchaye project served as an intensive crucible for advanced full-stack and cross-platform mobile development. We were immersed in a complex, multi-faceted technical environment, architecting and implementing a complete digital ecosystem comprising a robust backend API, web admin panels, and a citizen-focused mobile application and website, all designed to handle sensitive data and high-traffic loads during critical election periods.

Our core technical contributions and learnings included:

- Cross-Platform Mobile Development with Flutter: A central pillar of our work was the development of the citizen-facing mobile application using the Flutter framework. This involved:
    - Building a responsive and intuitive user interface (UI) with Dart and the Flutter widget library, ensuring a consistent experience on both iOS and Android devices for millions of potential users.
    - Implementing efficient state management solutions to handle dynamic content like real-time news feeds, candidate lists, and election updates.
    - API Integration: Architecting and coding the complete data layer of the app by integrating with our custom Laravel RESTful API. This involved crafting HTTP requests, parsing JSON responses, handling asynchronous data flows with async/await, and managing loading and error states gracefully.
    - Offline-First & Accessibility Design: A critical technical challenge was ensuring accessibility across Ethiopia's diverse network conditions. We implemented offline caching strategies to store essential election information (like candidate profiles and party manifests) locally on the device, allowing citizens to access crucial data without an active internet connection. Furthermore, we optimized data usage and app performance for low-bandwidth scenarios.

    - Privacy-Centric Authentication: Adhering to the project's core principle, we implemented the anonymous, no-login-required access model within the app flow, ensuring user privacy and removing barriers to access.
- Full-Stack Web Architecture & Development: We engineered the entire web-based backbone of the platform. This involved developing a secure RESTful API backend with Laravel (utilizing Eloquent ORM, MVC architecture, and Laravel Passport for authentication) and building dynamic, responsive admin dashboards for NEBE , political parties and individuals using the Blade templating engine and the Next.js React framework.
- Database Design & Optimization: We designed and implemented complex, normalized database schemas in MySQL to efficiently manage relationships between entities like users, political parties, candidates, news articles, and multilingual content. A significant achievement was the strategic implementation of data caching mechanisms on the backend, which resulted in a reduction in database load for high-frequency endpoints—a crucial optimization to ensure the mobile app and website would perform seamlessly under anticipated traffic surges.
- Security-First Development: Given the platform's sensitive nature, we implemented a multi-layered security strategy across both the backend API and the mobile app. This included role-based access control (RBAC) with granular permissions; secure API communication using tokens; IP-based rate limiting to mitigate abuse; and rigorous data sanitization to prevent injection attacks. The foundational "privacy-by-design" principle was enforced throughout the entire system.
- Deployment & DevOps Practices: We gained hands-on experience with the entire deployment lifecycle, from configuring staging environments in cPanel for the web backend to building and preparing release versions (APK and IPA) for the Flutter mobile application. This involved managing different environments, troubleshooting server and build configurations, and understanding the end-to-end process of taking software from development to production.
- Advanced Feature Implementation: The project challenged us to solve unique, real-world problems, such as building a robust multilingual content management system that served both the web and mobile apps, and creating an audit logging system for transparency.

3.1.2 Non-technical experience

Internship at the Ewenet Communication PLC as a software engineering student also Beyond the code, the internship was a masterclass in professional soft skills and understanding the broader context of technology's role in society.

- Stakeholder Engagement & Requirements Analysis: We learned to translate the needs of diverse stakeholders—from NEBE officials to political parties and rural citizens—into clear technical specifications. This involved analyzing existing electoral documents, and practicing empathetic listening to ensure the platform solved real, on-the-ground problems.
- Agile Project Management in Practice: Our work was governed by Scrum methodology, with weekly sprint planning and review sessions. This ingrained in us the disciplines of iterative development, task prioritization, adaptive planning, and the importance of regular feedback loops for continuous improvement.
- Interdisciplinary Collaboration & Communication: We worked as part of a cohesive team alongside a System Architect, UI/UX designers, and project managers. This required us to articulate complex technical challenges to non-technical team members, incorporate design feedback, and collaborate on problem-solving, sharpening our verbal and written communication skills.
- Ethical & Legal Considerations: Developing a tool for elections immersed us in critical ethical considerations: political neutrality, the fight against misinformation, data privacy, and digital equity. We developed a deep understanding of the ethical responsibility that comes with building civic technology and the importance of adhering to legal and regulatory frameworks.
- Ownership & Project Leadership: Taking ownership of major components of the backend and contributing to architectural decisions fostered a strong sense of responsibility and initiative. The experience of presenting our work to mentors and stakeholders built confidence and honed our ability to advocate for and justify our technical decisions.
- Problem-Solving Under Constraints: We regularly navigated complex challenges, such as optimizing for limited bandwidth, resolving deployment issues, and refining features based on user feedback. This enhanced our critical thinking and resilience, teaching us to find innovative solutions within technical, time, and resource constraints.

3.2 Benefits of experience

The benefits of working on the Mirchaye platform extend far beyond the duration of the internship, providing a powerful foundation for our future careers.

- Portfolio of Tangible Impact: We contributed to a live, nationally significant platform that will directly strengthen democratic engagement in Ethiopia. This demonstrable experience is a powerful differentiator that showcases our ability to deliver complex, mission-critical software.
- Holistic Skill Advancement: The internship provided a rare opportunity to simultaneously advance deep technical expertise in modern web development stacks and cultivate essential professional soft skills like project management, client communication, and ethical reasoning.
- Understanding of Security Protocols: Interns working with Ethiopian Election information systems gain insights into legal requirements , security measures, data encryption, access control, and compliance with safety regulations. This knowledge can be applied to various fields, making interns well rounded professionals.
- Collaboration: Interns work alongside experienced professionals in a team environment. This fosters collaboration, communication, and teamwork skills, which are vital in any professional setting.
- Networking Opportunities: Internships provide a unique platform for building professional connections. Interns have the opportunity to interact with professionals in the field, which can be advantageous for future career prospects.
- Resume Building: Experience in Ethiopian Election information systems can significantly enhance an intern's resume. It demonstrates practical expertise in a specialized area and can make them more attractive candidates for future job opportunities in political, law enforcement, or related fields.
- Problem-Solving: Interns often encounter real-world challenges and complexities related to Mirchaye Election information . This practical experience encourages them to develop effective problem-solving skills.
- Project Management: Exposure to project management processes and tools is a common aspect of internships. Interns may learn how to plan, execute, and monitor projects effectively.

- Project Ownership: In some cases, interns may take ownership of specific projects. This responsibility allows them to see the direct impact of their work and take initiative in achieving project goals.
- Increased Confidence: Successfully contributing to a specialized field like Ethiopian Election information society can boost interns' confidence in their abilities and decision-making.
- Personal Growth: Dealing with complex and sensitive issues in Ethiopian Election Information can lead to personal growth. Interns may develop a heightened sense of responsibility, ethical judgment, and a more profound understanding of the work they are involved in.

3.3 Skills you get from experience (soft and hard skills)

Hard Skills:

- Full-Stack Development Proficiency: Gained extensive, hands-on experience in both front-end and back-end development, specifically mastering the Laravel PHP framework (Eloquent ORM, MVC architecture, Blade templating, API development with Laravel Passport) and building dynamic user interfaces with Next.js and React.
- Cross-Platform Mobile Development: Acquired expertise in developing a production-ready mobile application using the Flutter framework, including UI design with widgets, state management, and integrating with a custom RESTful API.
- Database Design & Management: Developed advanced skills in relational database design with MySQL, including schema normalization, writing complex queries, optimizing performance through indexing, and implementing caching strategies to reduce server load .
- API Architecture & Integration: Learned to design, build, document, and consume secure RESTful APIs, serving as the critical bridge between the Laravel backend, the Next.js admin panels, and the Flutter mobile application.

- Version Control & Collaboration with Git: Mastered professional Git workflows in a team environment, including branching strategies, committing, pushing, creating pull requests, and conducting code reviews on platforms like GitHub.
- Deployment & DevOps Fundamentals: Gained practical knowledge of deploying a full-stack application to a live cPanel environment, configuring servers, managing file permissions, migrating databases, and resolving production-level issues.
- Security Implementation: Understood and applied critical security principles, including role-based access control (RBAC), data validation/sanitization, API rate limiting, and building systems with a "privacy-by-design" approach.
- Multilingual System Design: Learned the technical architecture behind implementing a content management system that efficiently supports multiple local languages, a key feature for national accessibility.

Soft Skills:

- Agile Methodology & Scrum Execution: Practiced Agile development firsthand through weekly sprint planning, task estimation, and iterative development, enhancing adaptability and project management capabilities.
- Stakeholder Communication: Improved the ability to translate complex technical concepts into clear language for non-technical stakeholders, including project managers and designers, and to understand their requirements and feedback.
- Problem-Solving & Critical Thinking: Strengthened analytical skills by constantly diagnosing and resolving complex technical challenges, from debugging intricate code errors to architecting solutions for scalability and offline access.
- Team Collaboration & Interdisciplinary Work: Learned to work effectively within a team of diverse roles—backend developers, frontend developers, UI/UX designers, and project managers—fostering teamwork and mutual respect.
- Time Management & Task Prioritization: Honed the ability to manage time efficiently across a 16-week internship, balancing learning new technologies, meeting weekly development goals, and contributing to a large-scale project deadline.

- Attention to Detail: Cultivated a meticulous eye for detail, crucial for writing clean code, catching bugs, ensuring UI consistency, and maintaining data integrity across a system handling critical election information.
- Ownership & Initiative: Developed a sense of professional responsibility by taking ownership of assigned modules, proactively seeking solutions to blockers, and driving tasks to completion without constant supervision.
- Adaptability & Continuous Learning: Demonstrated the ability to quickly learn and apply new frameworks and technologies (Laravel, Flutter, Next.js) as required by the project's evolving needs.

3.4 Challenges you face and Solution you have taken (personal and company challenges)

The development of a nationally significant platform like Mirchaye presented a unique set of personal and company-wide challenges that required innovative and resilient solutions.

Personal Challenges

- Steep Learning Curve in New Technologies: The requirement to quickly become proficient in unfamiliar but critical technologies like Laravel and Flutter was initially daunting.

  - Solution: Dedicated significant personal time outside of work hours to online tutorials, official documentation, and building small practice projects. Paired programming with a more experienced peer also accelerated the learning process.

- Understanding Complex Domain Logic: Grasping the intricacies of Ethiopia's electoral processes, legal requirements, and the specific workflows of NEBE officials was complex and outside standard software knowledge.

  - Solution: Conducted intensive research and asked numerous questions during stakeholder interviews. Created detailed flowcharts and documentation to map out processes before writing any code, ensuring the software model matched the real-world domain.

- Debugging Complex, Interconnected Systems: Troubleshooting errors that occurred at the intersection of different systems (e.g., an API response not rendering correctly in the Flutter app) was particularly challenging.

  - Solution: Learned to use debugging tools systematically, starting with the backend API tests (PHPUnit, Postman), then checking the network requests from the frontend, and finally isolating the issue in the mobile or web client code.

- Managing Work-Life Balance During Intensive Development: The demanding nature of the project, especially during the crucial weeks leading to deployment, required managing stress and maintaining productivity.

  -Solution: Practiced time-blocking for focused work, took short breaks to avoid burnout, and maintained clear communication with the team about progress and potential bottlenecks.

Company Challenges

- Evolving Project Requirements: Initial requirements from stakeholders evolved as the project progressed and they better understood the platform's potential, leading to scope adjustments.

  -Solution: Embraced the Agile mindset. The team held feedback sessions to reassess priorities and adjust the sprint backlog accordingly, ensuring the product evolved to meet the most critical user needs without derailing the entire timeline.

- Stakeholder Availability for Requirement Gathering: Key professionals within Ewenet had extremely busy schedules, making it difficult to secure time for in-depth interviews and feedback.

  -Solution: Maximized every scheduled meeting by preparing extremely specific, targeted questions in advance. We also used asynchronous communication (email, shared documents) to present mockups and proposals for review at their convenience.

- Technical Constraints and Environmental Disparities: Ensuring the application would perform reliably for users across Ethiopia, including those with low-end devices and poor internet connectivity, was a major technical hurdle.

  - Solution: Designed and implemented an "offline-first" approach for the mobile app, caching essential data locally. Optimized images and API payloads for size and implemented lazy loading to minimize data usage.

- Ensuring Data Security and Political Neutrality: The sensitive nature of election data and the absolute necessity for the platform to be perceived as neutral imposed a high burden of care.

  - Solution: Instituted a multi-layered security review process and implemented strict role-based permissions. All content from political parties was clearly labeled as such, and a transparent, auditable logging system was built to ensure all actions on the platform could be trace

4 General Overview of the project

The objective of Mirchaye Platform is to address critical challenges in Ethiopia's electoral ecosystem, including widespread misinformation, fragmented information channels, and limited access to credible election-related content. The platform aims to provide a centralized, accessible, and politically neutral hub for real-time election information, political party profile and individual profiles,who may be running for seats in the House of People's Representatives or other election-related roles, and National Election Board of Ethiopia (NEBE) announcements. By leveraging mobile apps, web interfaces, and secure admin panels, the Mirchaye Platform ensures anonymous access to multilingual content, fostering transparency, combating misinformation, and empowering voters to make informed decisions.

4.1 Background of the project

The background of the Mirchaye Platform is rooted in the imperative of electoral transparency, public trust, and democratic engagement. The National Election Board of Ethiopia (NEBE) and other democratic institutions are entrusted with ensuring fair and informed electoral processes, necessitating a robust system to ensure the meticulous management and dissemination of election information. This platform is designed to enhance civic safety by reducing the risk of misinformation and ensuring citizens have access to lawful, verified electoral content. By transitioning from fragmented, often manual information channels to a centralized automated system, the efficiency, accuracy, and accountability of public communication are significantly improved. Furthermore, a comprehensive electoral information system provides real-time news updates, multilingual reporting capabilities, and privacy-focused security measures, ensuring that NEBE, political parties, and individual candidates can maintain a high level of transparency and operational integrity. The Mirchaye Platform plays a pivotal role in maintaining the integrity and inclusivity of Ethiopia's democratic process by enabling meticulous verification of information, compliance with electoral guidelines, and efficient, equitable access to information for all citizens.

4.2 Objective of the project

4.2.1 General objective

The general objective of the project is to transition the existing fragmented and often manual election information ecosystem in Ethiopia into a modern, automated, and centralized digital platform. This transformation is essential to enhance the efficiency, accuracy, and overall effectiveness of disseminating electoral information, including news, party and candidate profiles, and official announcements, which are crucial to democratic engagement and public trust.

The current information dissemination system presents several inherent challenges. It relies heavily on disparate channels, including social media, traditional media, and physical documentation, making it prone to misinformation, delays, and inconsistency. Additionally, the accessibility and verification of credible information can be cumbersome and time-consuming, impacting the ability of citizens to make informed decisions and undermining the integrity of the electoral process.

The transition to an automated web and mobile-based platform addresses these challenges comprehensively. It involves the development of a software solution that streamlines the entire electoral information lifecycle, from creation and verification by authoritative sources (NEBE, parties, candidates) to distribution and consumption by the public.

4.2.2 Specific objective

The specific objectives required to achieve the general goal of automating the electoral information system for the National Election Board of Ethiopia and the Ethiopian public are as follows:

- Interactive and Accessible User Interface: Develop an interactive, user-friendly, and multilingual interface that ensures easy accessibility and usability for all citizens, regardless of technical literacy or language.

- Centralized Database Creation: Create a secure database that will serve as the central repository for storing election news, political party profiles, individual candidate information, and user access data, ensuring organized and integrity-rich data management.

- Cross-Platform System Deployment: Successfully deploy the web-based admin panels and mobile applications, making them accessible to all users and ensuring high availability and reliability, especially during critical election periods.

- Political Party and Candidate Management: Establish a streamlined management system for political party and individual candidate profiles and content, enhancing the transparency, accuracy, and control of information presented to the public.

- NEBE Announcement and News Management: Develop a robust management system for official NEBE announcements, electoral news, and educational content, improving the speed and reliability of distributing critical updates to the electorate.

- Privacy and Security Enhancement: Implement a privacy-first design that removes the need for user registration for accessing public content, ensuring anonymity and protecting users from potential surveillance, thereby fostering greater trust and engagement.

4.3 Statement of the problem for the project

The Ethiopian electoral information ecosystem currently operates through fragmented and inefficient channels, leading to significant obstacles in public engagement and transparency. The core problems faced are: -

- Pervasive Misinformation and Propaganda: The dominance of social media and informal networks for information sharing leads to the rapid spread of unverified claims and biased narratives, making it difficult for citizens to distinguish fact from fiction.

- Lack of a Centralized Information Source: Official electoral updates, party profiles, and candidate information are scattered across various non-digital and online platforms. This forces citizens to seek information from multiple, often unreliable, sources.

- Digital Privacy Concerns: Many existing digital platforms require user registration or collect personal data. In a politically sensitive climate, this deters citizen engagement due to fears of surveillance and tracking.

- Cumbersome Voter Education: The current system makes it challenging for voters to easily compare political party agendas or individual candidate profiles, hindering their ability to make informed decisions.

- Difficulty in Ensuring Information Authenticity: The manual and distributed nature of the current system makes it nearly impossible to quickly verify the accuracy of electoral information and counter false narrative

4.4 Methodology

The following methodologies were employed to gather requirements and guide the development of the Mirchaye Platform:

- Document analysis: We analyzed existing NEBE publications, political party manifestos, campaign materials, and official communication models used for public announcements. This helped define the structure and type of content required for the platform.

- Observation: We observed the current electoral information landscape, noting how long it takes for citizens in different regions to receive official updates and the challenges they face in verifying news from social media and other informal channels.

- Interview: This was a primary methodology for gathering user requirements. We interviewed:

    - NEBE Officials: To understand official communication protocols and legal frameworks.

    - Political Party Representatives: To identify their needs for public engagement and content management.

    - Media Experts and Civil Society Analysts: To gather insights on combating misinformation and ensuring platform neutrality.

    - Potential Users (Citizens): To understand their needs for accessible, trustworthy, and private access to information.

    - **Significance of the Project**

The concept of having a single, neutral platform responsible for the entire flow of electoral information—from source (NEBE, parties, candidates) to consumer (the citizen)—is transformative for Ethiopia's democracy. If Ethiopia wishes to minimize the societal costs of misinformation and provide a higher standard of electoral transparency, transitioning to this digital system is essential.

- Fast information dissemination and retrieval.
- The new system provides service 24/7.
- It offers speed and efficiency in accessing information.
- Ensures accuracy through source verification.
- Reliance on modern, scalable technology.
- Data is secure and integrity is maintained.
- Can store a large amount of multilingual data and media content.
- Quick processing of information requests and searches.
- Provides a competitive advantage for informed voting.
- Reduces reliance on paper-based campaign materials and flyers.

➢ Scope of the documentation

This document covers the requirements, analysis, and design of the Mirchaye Platform. The scope includes the development and implementation of the core subsystems with the functionalities discussed in the table below. Additional functionalities may be added in future development phases as needed.

Table 1: Scope of the document

Number	Subsystem	Functionalities
1	News & Announcement Management	Insertion, updating, and deletion of election news and official announcements.
2	Profile Management	Managing the profiles and content of political parties and individual candidates.
3	Content Distribution	Managing the distribution of content to citizens via the app, web, and push notifications.
4	Auditing or Logging	Everyone has the responsibility of its actions.

➢ Scope of the project

The Mirchaye Platform will work on the following scope:

- To allow NEBE admins to post and manage official news.
- To allow political party admins to create and manage their profiles and posts.
- To allow individual candidate admins to create and manage their profiles and posts.
- To display verified election news and profiles to users.
- To enable users to search for parties, candidates, and news.
- To provide content in multiple local languages.
- To cache content for offline viewing.
- To generate audit or logging for admin use.

➢ Limitation of the project

- Our system does not include live streaming of campaign events or debates.
- User identity verification for advanced features (e.g., online voting) is out of scope.
- When there is no internet connection, real-time updates cannot be received, though cached content remains available.
- Our system does not directly integrate with external voter registration databases.

4.5 Development tool

Table 2: Tools used by project

Number	Component	Technology/Tool
1	Backend API	Laravel (PHP)
2	Admin Panel	Laravel Blade
3	Mobile Frontend	Flutter
4	Web Frontend	Next.js
5	Database	MySQL
6	Version Control	Git / GitHub
7	Code Editor	Visual Studio Code
8	Hosting	cPanel

➢ REQUIREMENT ANALYSIS

Introduction

It is possible to develop a better system by understanding the existing electoral information ecosystem. Even in its current fragmented state, the process of sharing election information performs the following activities: The National Election Board of Ethiopia (NEBE) releases official statements and announcements, political parties and individual candidates campaign to share their agendas with the public, and citizens seek out this information to make informed voting decisions.

➢ Description of the existing System

The existing electoral information system performs its function through a fragmented, manual, and digital hybrid system, which leads to significant issues with misinformation, accessibility, and trust. Because of this disjointed system, the process of finding, verifying, and sharing credible election information is time-consuming and inefficient. This is the result of a lack of a centralized, computerized system. The current ecosystem deals with the following major tasks: namely, news generation (NEBE, parties, media), information distribution, and citizen consumption.

Work flow of the existing system

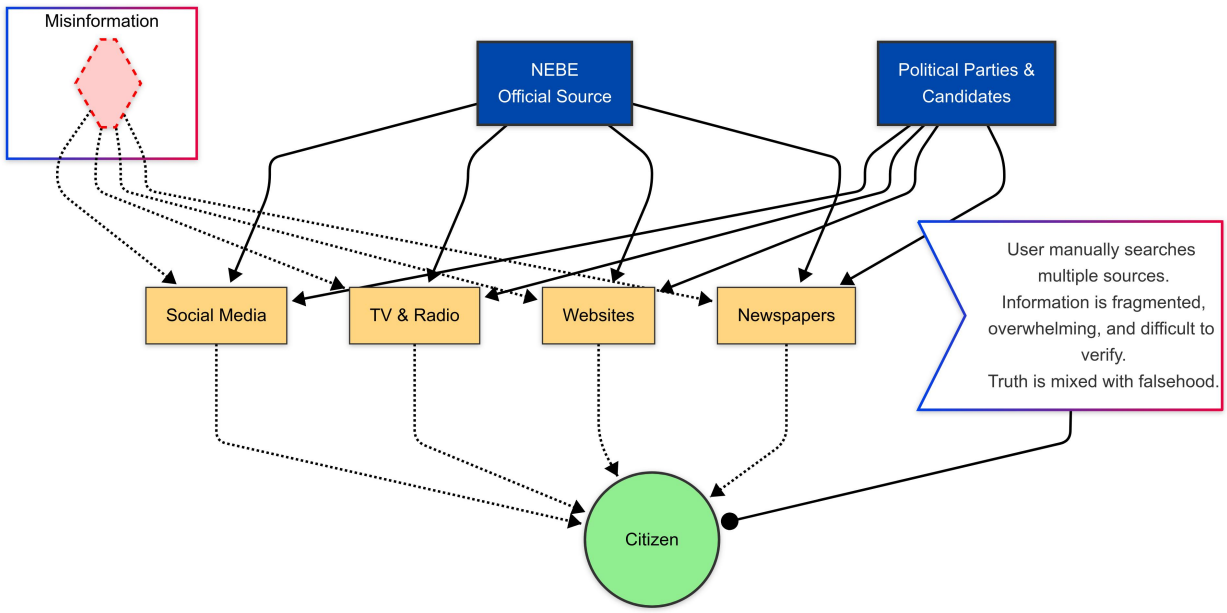


Figure 1 : work flow of Existing System

➢ **The existing system**

The existing system requires NEBE, political parties, and candidates to prepare their announcements and campaign materials manually and distribute them across numerous disconnected platforms (social media, websites, pamphlets, TV). This manual distribution is inefficient and consumes excessive time and human resources. Furthermore, the information generation and distribution process is not automated, meaning that when a citizen wants to find comprehensive information, they cannot get it automatically and in a detailed, comparable format from a single source. In general, the existing information distribution method is ineffective. Because of this, the existing system must be automated into the new system, which is the Mirchaye Platform.

➢ **Proposed system**

This is the new system that removes the existing problems which usually occurred in the fragmented information ecosystem. The main purpose of the proposed system is to improve the existing system, simplify the workload for all stakeholders, and speed up the operation of the overall electoral process. The application areas for the proposed system have been selected on the basis of the following factors:

- Minimizing the Information Silos: Centralizing information kept in different locations and platforms.
- There will be more data integrity and trust: All content is verified at the source.
    - NEBE will distribute official news.
    - Parties/Individual Candidates will view and manage their own distributed content.
    - Citizens will view the distributed information.
- Facilitate desired information display very quickly by retrieving information from a single, authoritative API.
- Facilitating various statistical information which helps in decision-making for parties and NEBE (e.g., engagement metrics).
- To reduce manual efforts in activities that involve repetitive work like distributing the same announcement across multiple channels.

➢ **Specific requirement**

Requirement specification is an agreement between the end user and the system developer; therefore, this requirement specification is the agreement made with the Mirchaye Platform stakeholders (NEBE, Political Parties, Individual Candidates, and Citizens) and the system developer.

User requirements are a document that defines what a proposed system must be capable of doing to solve the existing problems. These requirements are classified into functional requirements and non-functional requirements, which are detailed in sections 4.6 and 4.7 of this document.

4.6 Functional Requirements

FR-1: The system shall have precreated NEBE admin account.

FR-2: The system shall register political parties and individual candidates with their respective information in order to create account and await for NEBE approval.

FR-3: The system shall not require registration or login from end-users (citizens) to access content.

FR-4: The system shall provide role-based access control (e.g., NEBE admin, party admin, candidate admin).

FR-5: The system shall be fully responsive and mobile-friendly.

FR-6: The system shall be accessible in Amharic, Afaan Oromoo, English, Tigrinya, and Somali.

FR-7: The system shall allow offline access to previously loaded news, party, and candidate profiles.

FR-8: The system shall allow users to filter news by category.

FR-9: The system shall allow users to search for political parties and individual candidates by name.

FR-10: The system shall provide search suggestions as users type.

FR-11: The system shall display real-time election news and official updates from NEBE.

FR-12: The system shall display verified public profiles of registered political parties, individual candidates and NEBE.

FR-13: The system shall prevent unauthorized access to political party, individual candidates and NEBE admin functions.

FR-14: The system shall allow political parties,individual candidates and NEBE to post content and event in multiple languages.

FR-15: The system shall allow NEBE administrators to publish, update, and delete announcements.

FR-16: The system shall support full CRUD (Create, Read, Update, Delete) operations for post and event with secure access for NEBE admins.

FR-17: The system shall support full CRUD (Create, Read, Update, Delete) operations for education material with secure access for NEBE admins

FR-18: The system must enable NEBE administrators to review, approve, or reject profile submissions from political parties and individual candidates.

FR-19: The NEBE administrators shall maintain secure audit logs of all administrative activities (every users are responsible for its action in our system).

FR-20: The system must allow political parties to create and manage their public profiles (biography, photo, agenda…).

FR-21: The system shall support full CRUD (Create, Read, Update, Delete) operations for campaign content with secure access for party admins.

FR-22: The system shall support full CRUD (Create, Read, Update, Delete) operations for event with secure access for party admins.

FR-23: The system must allow individual candidates to create and manage their public profiles (biography, photo, agenda…).

FR-24: The system shall support full CRUD (Create, Read, Update, Delete) operations for campaign content with secure access for individual candidates admins.

FR-25: The system shall support full CRUD (Create, Read, Update, Delete) operations for event with secure access for individual candidates admins.

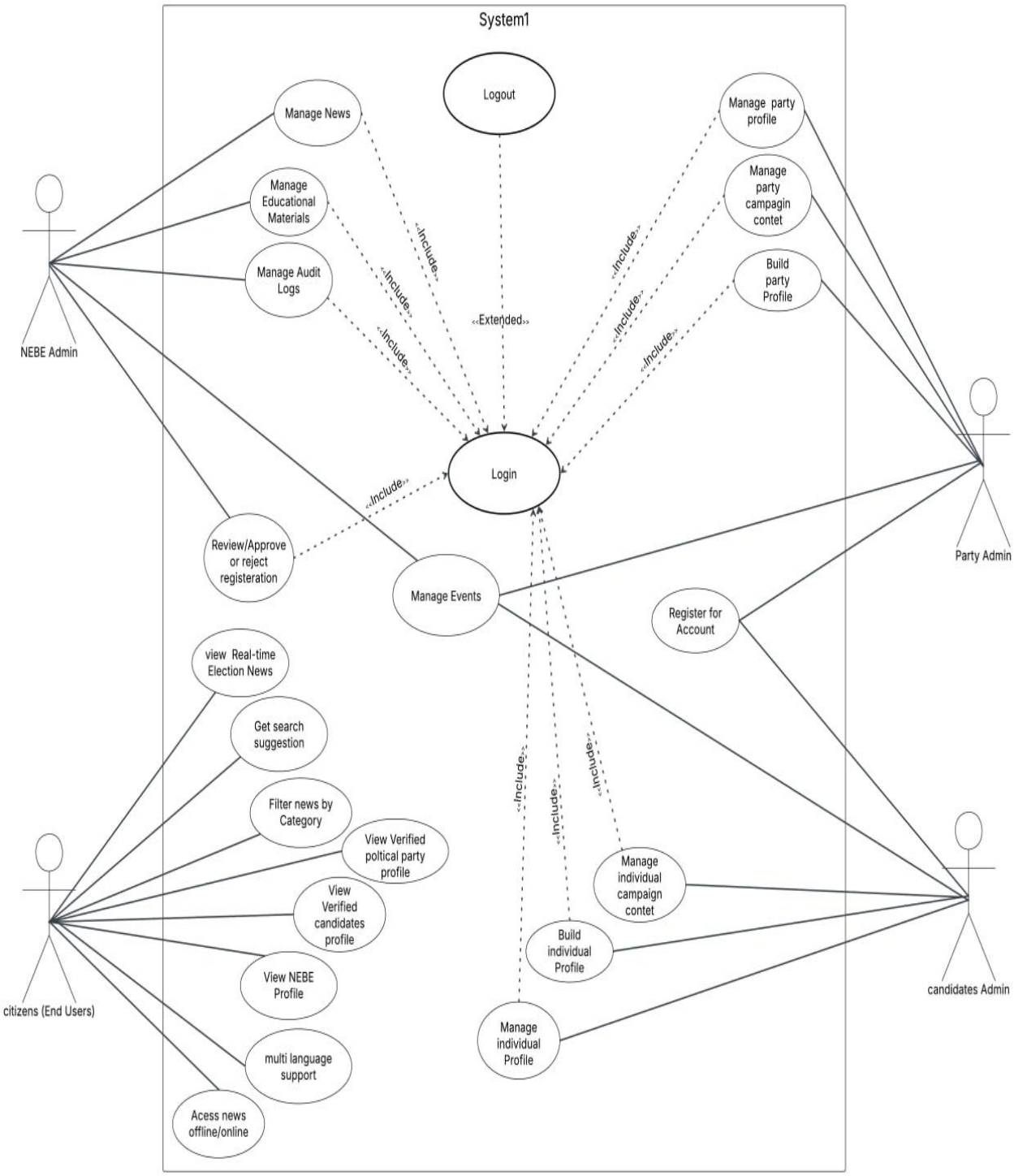
4.6.1 Non-Functional Requirements:

Table 3: Non-Functional Requirement

ID	Requirement	Description	Source	Priority
NF-REQ-1	Usability	The system shall provide an intuitive, user-friendly interface with a help menu, accessible to non-technical users.	Stakeholder	High
NF-REQ-2	Security	The system must provide secure login for admins and use HTTPS/TLS to encrypt all data in transit.	Developer	High
NF-REQ-3	Performance	The system shall load the home screen within 2 seconds on a 3G mobile network.	Developer	High
NF-REQ-4	Availability	The platform shall maintain 99.5% uptime, especially during national election periods.	Stakeholder	High
NF-REQ-5	Reliability	The application shall recover automatically from minor network interruptions without crashing.	Developer	High
NF-REQ-6	Localization	The system must fully support UI and content in Amharic, Afaan Oromoo, English, Tigrinya, and Somali.	Stakeholder	High
NF-REQ-7	Privacy	The system shall allow full anonymous access without requiring registration or collecting PII.	Stakeholder	High
NF-REQ-8	Scalability	The system shall support up to 10,000 concurrent users without performance degradation.	Developer	High
NF-REQ-9	Maintainability	The codebase shall be modular and well-documented for ease of future updates.	Developer	Medium
NF-REQ-10	Auditability	All admin activities shall be logged with user ID, timestamp, and action details for audit purposes.	Stakeholder	Medium
NF-REQ-11	Portability	The system shall be deployable on Android, iOS devices, and modern web browsers.	Developer	Medium
NF-REQ-12	Data Integrity	The system shall ensure that all published party and candidate profiles are verified and tamper-proof.	Stakeholder	High

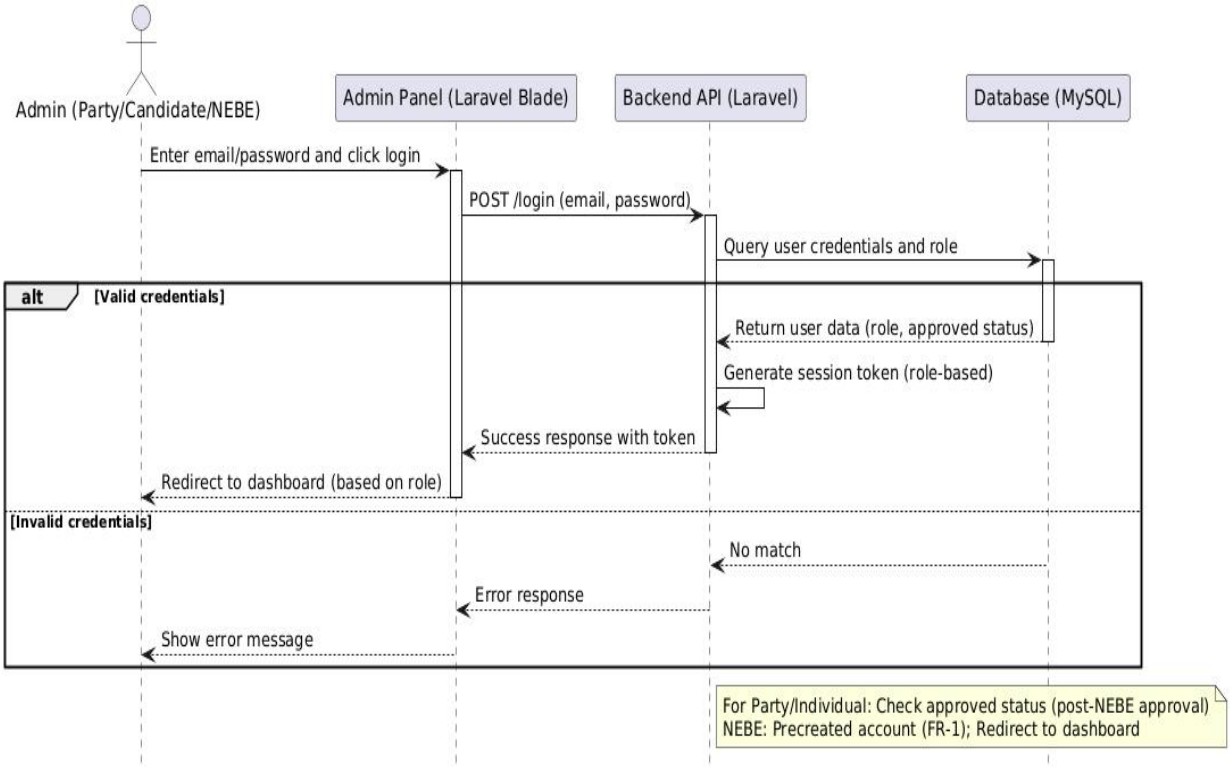
4.7 Diagram

4.7.1 Usecase diagram

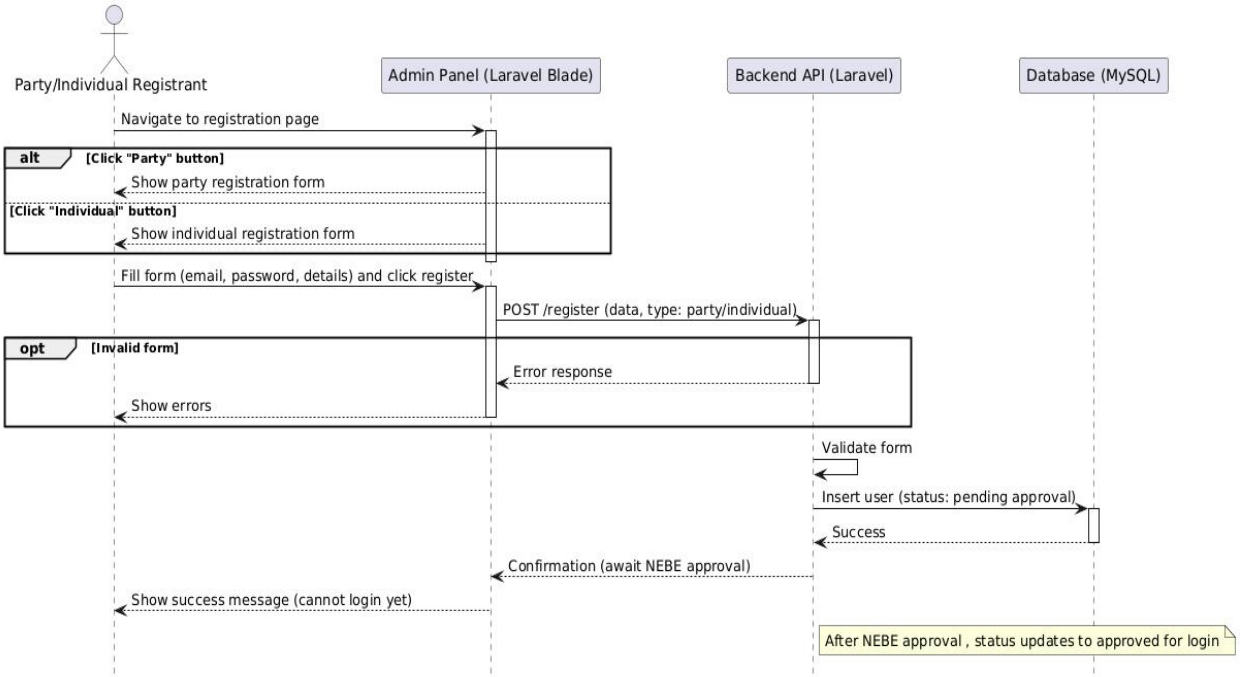


4.7.2 Sequence Diagram

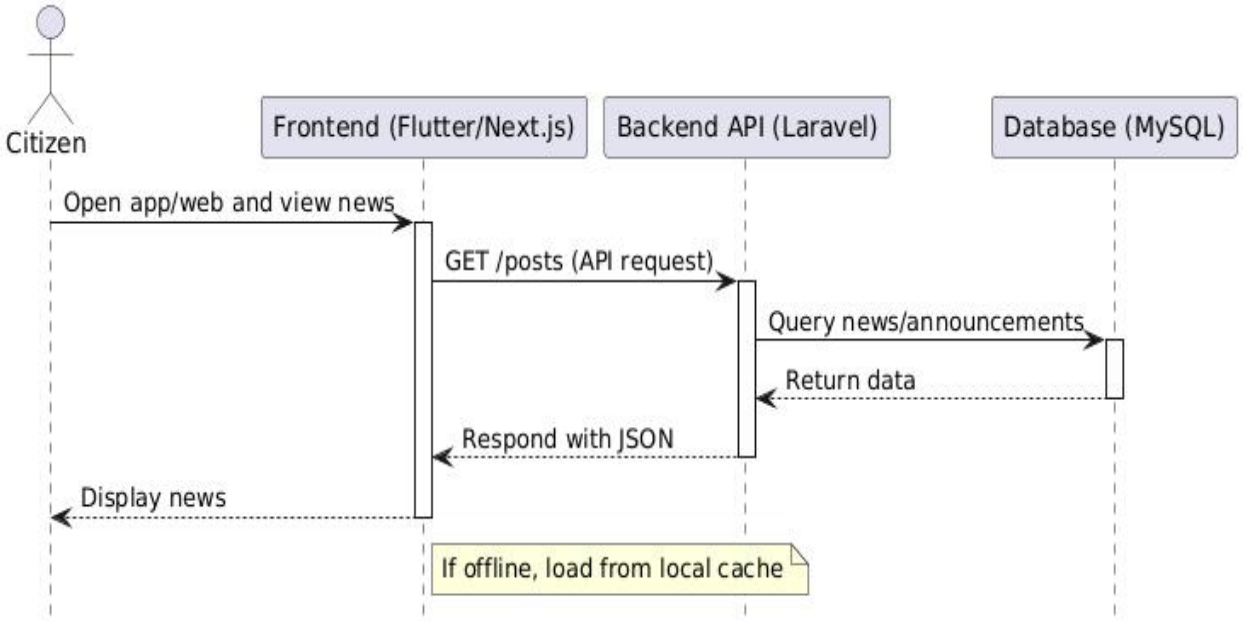
Sequence diagram for login



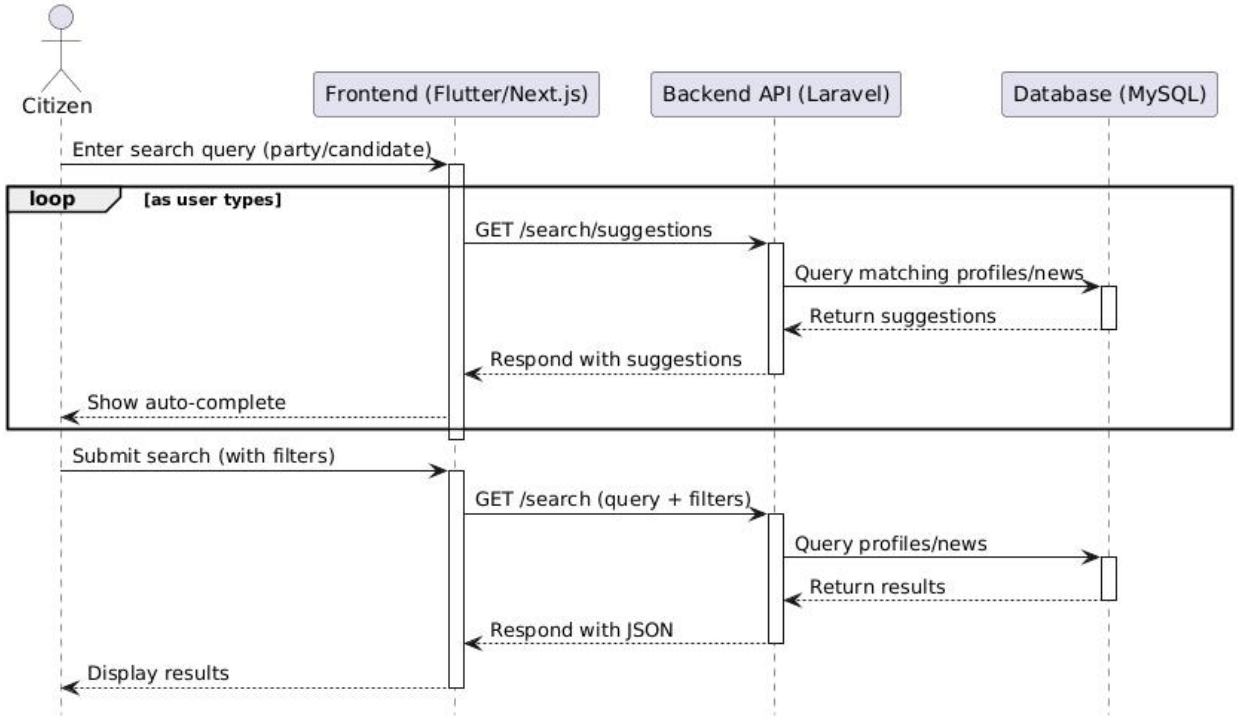
Sequence diagram for register



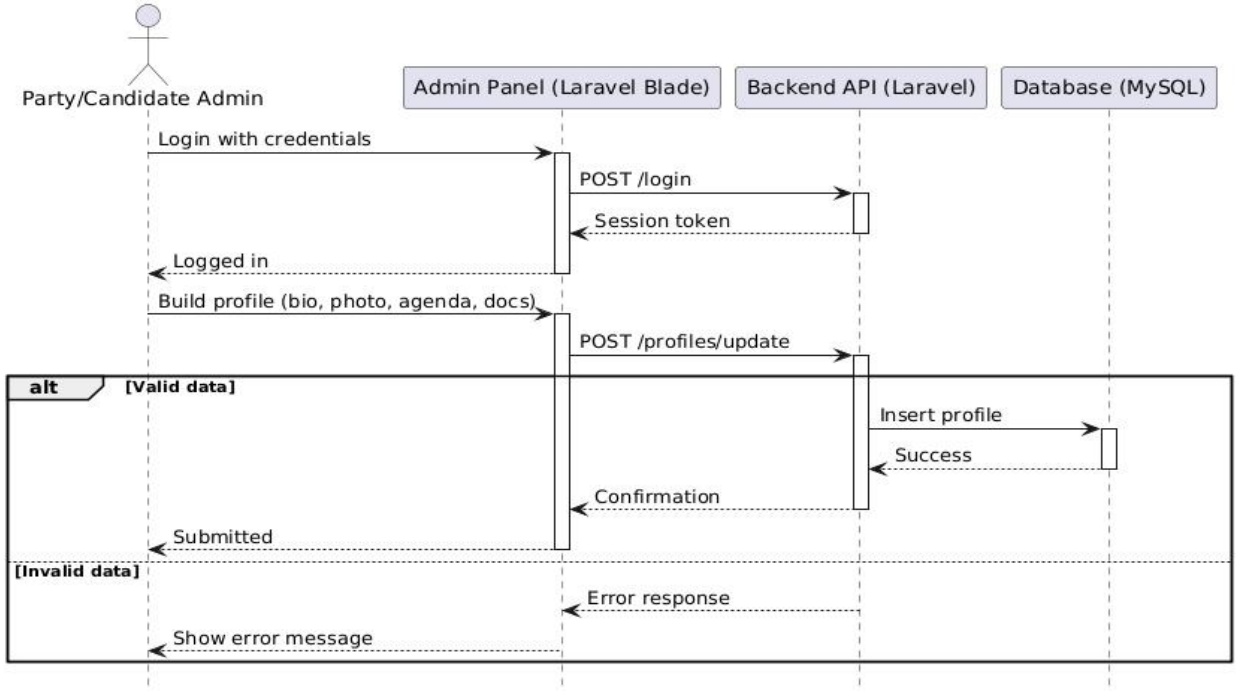
Sequence diagram for Citizen Viewing Election News



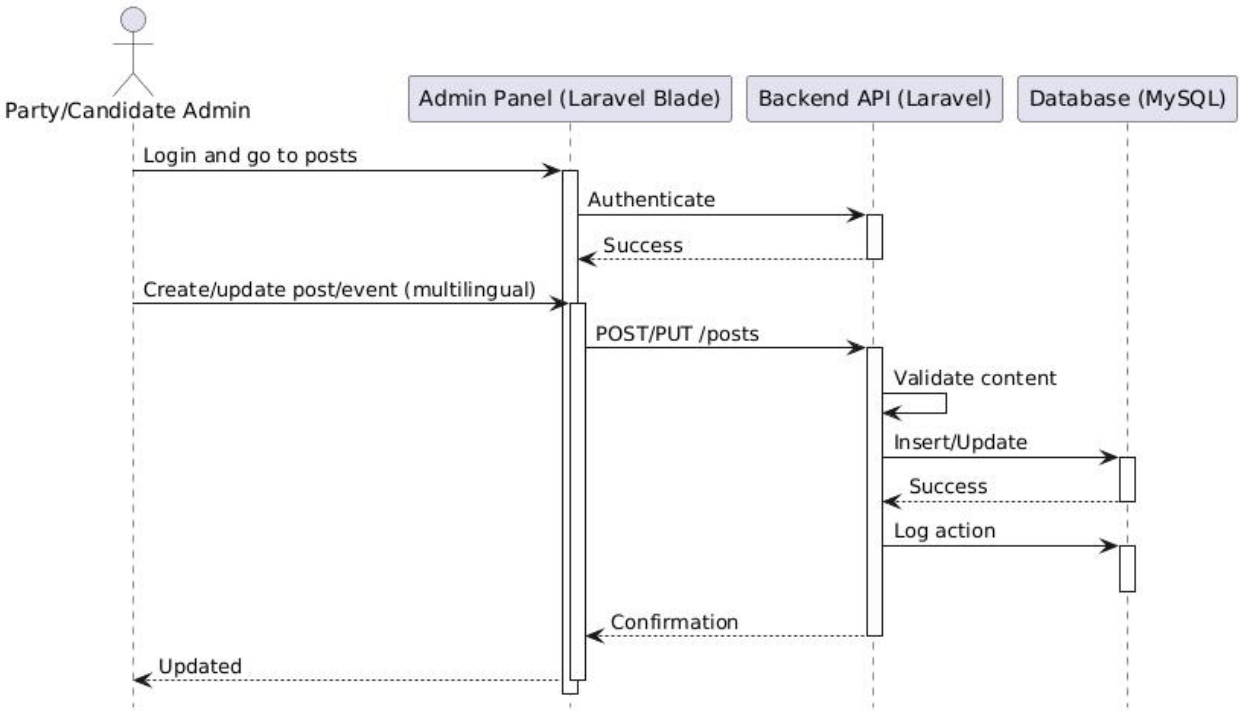
Sequence Diagram for Citizen Searching for Parties/Candidates



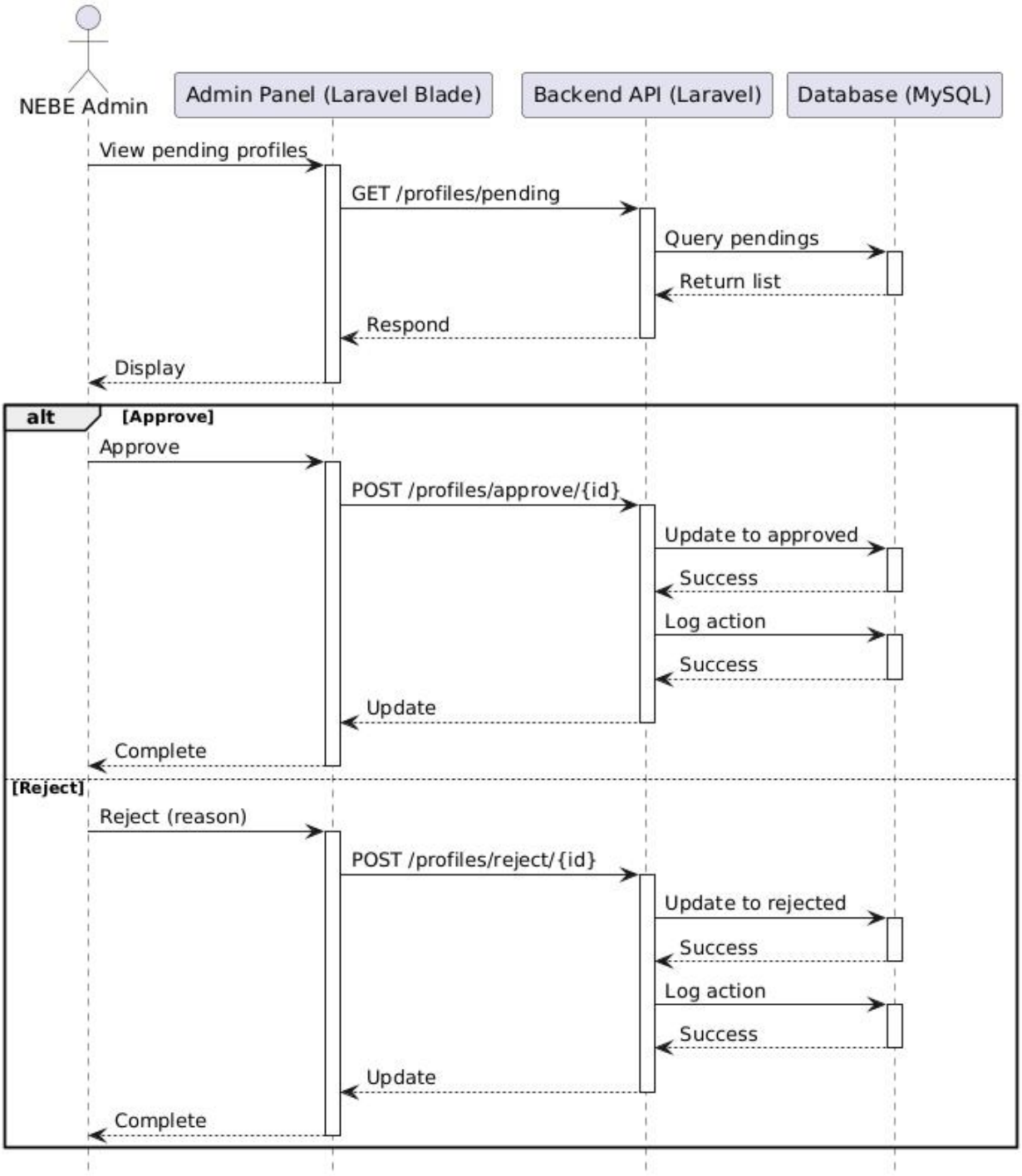
Sequence diagram for Party/Individual Candidate Building/Updating Profile



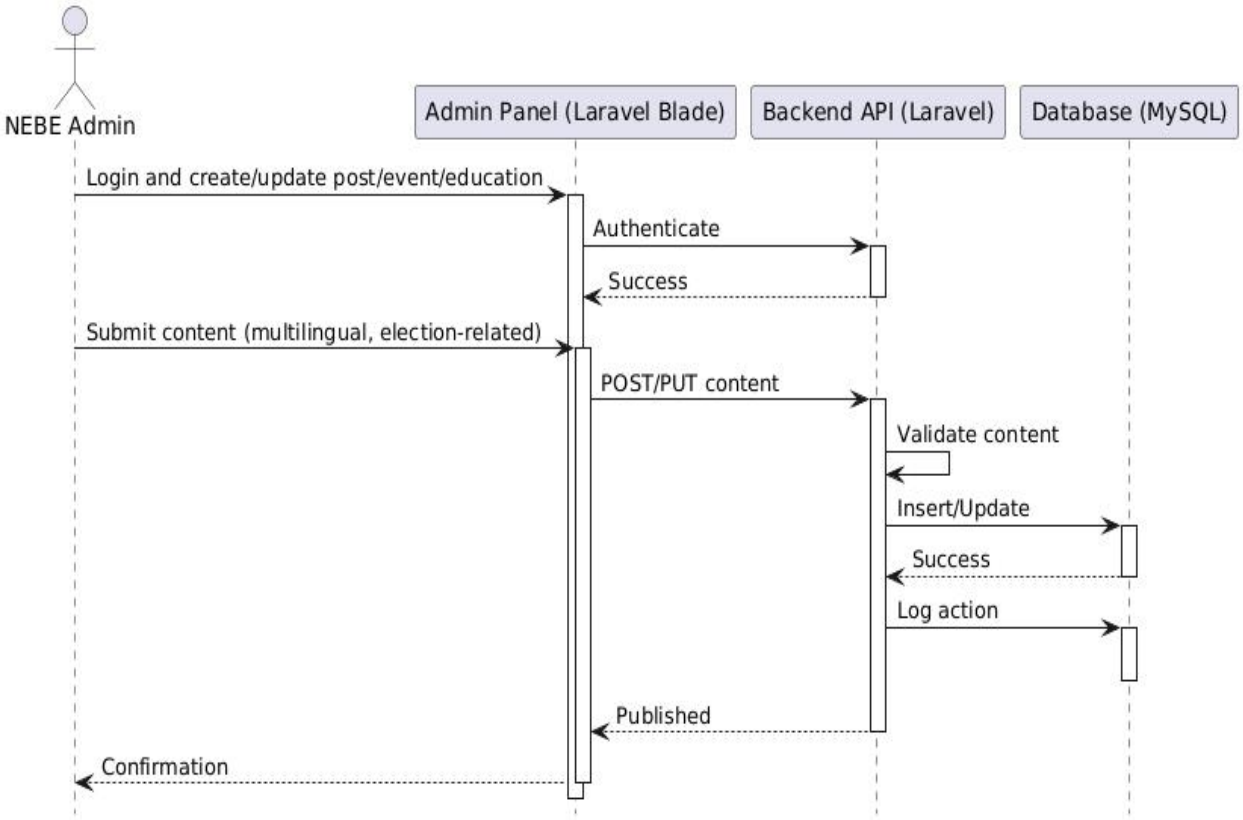
Sequence diagram for Party/Individual Candidate Creating/Updating Post/Event



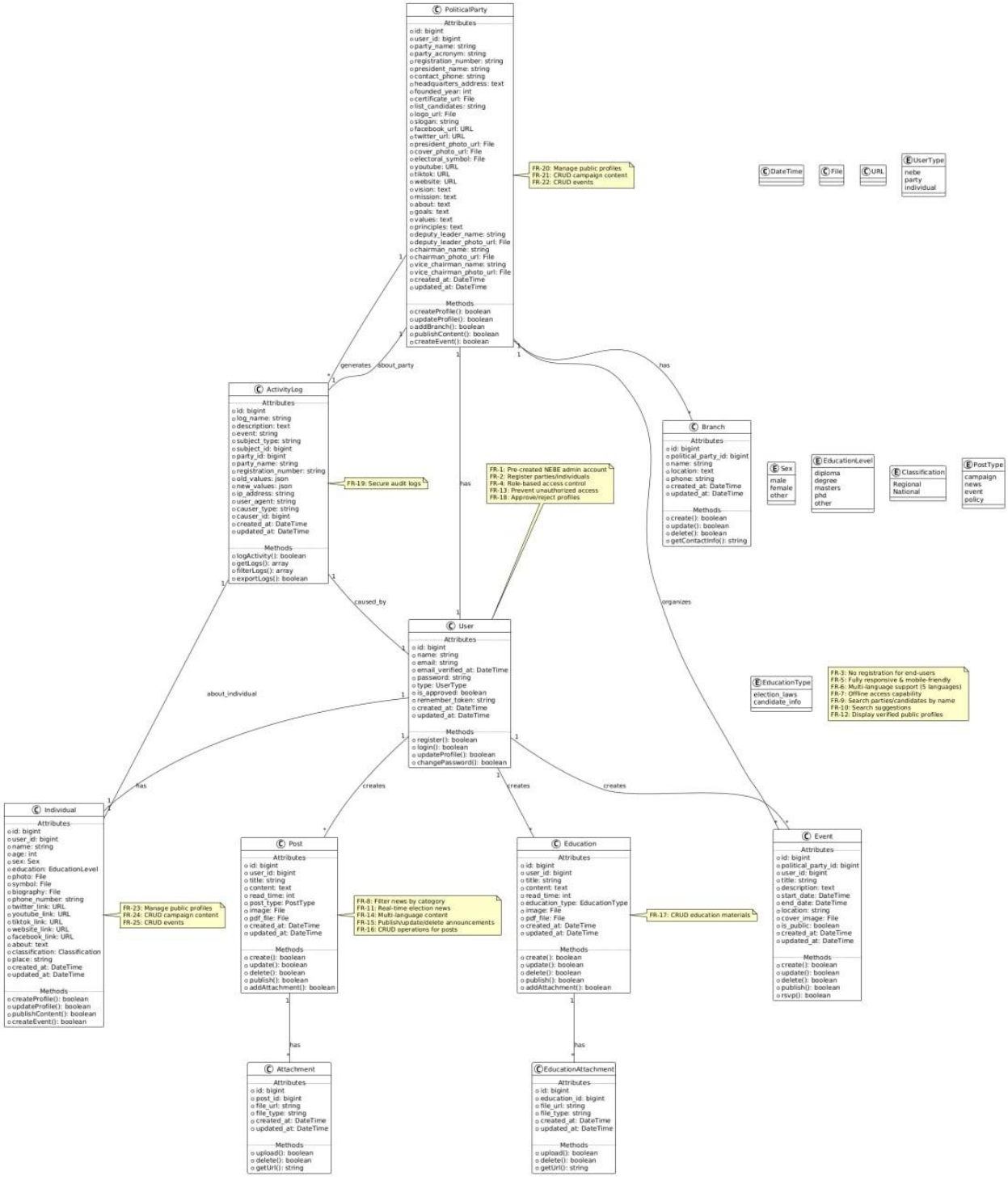
Sequence diagram for NEBE Approving/Rejecting Profile



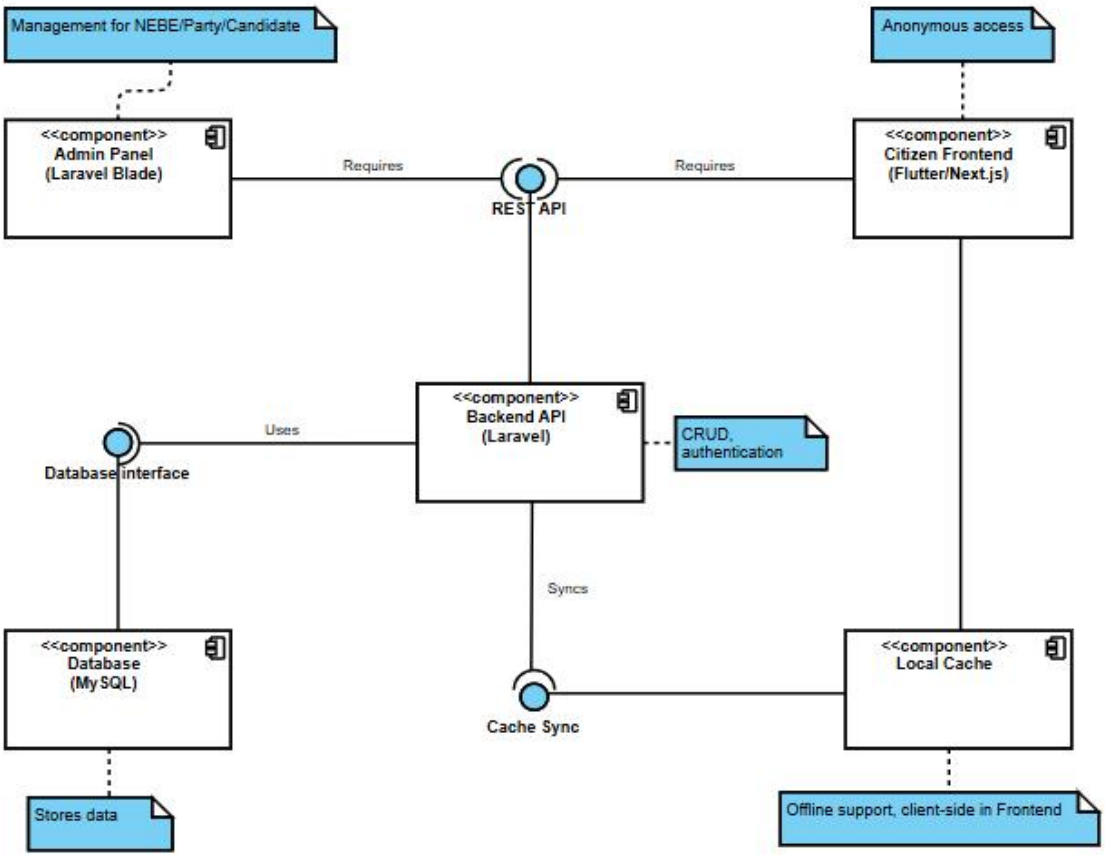
Sequence Diagram for NEBE Creating/Updating Post/Event/Education



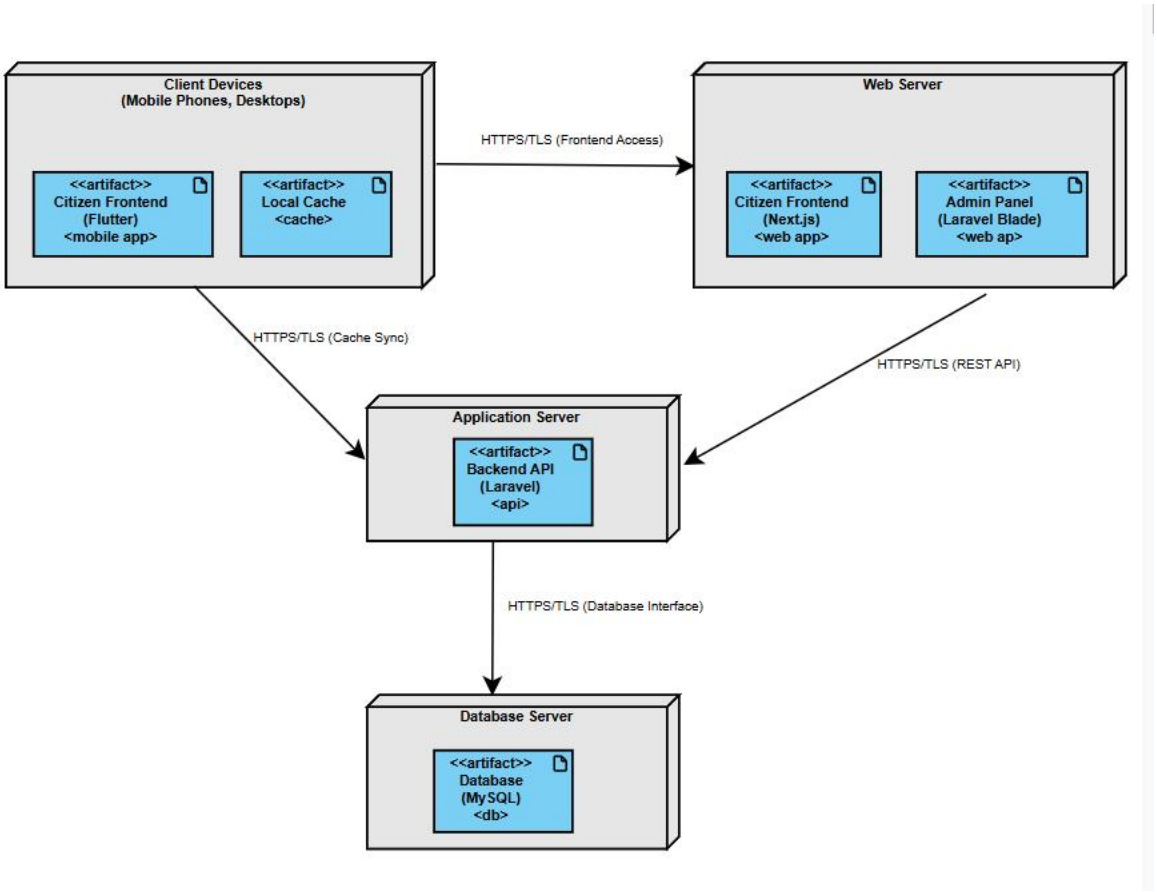
4.7.3 Class Diagram



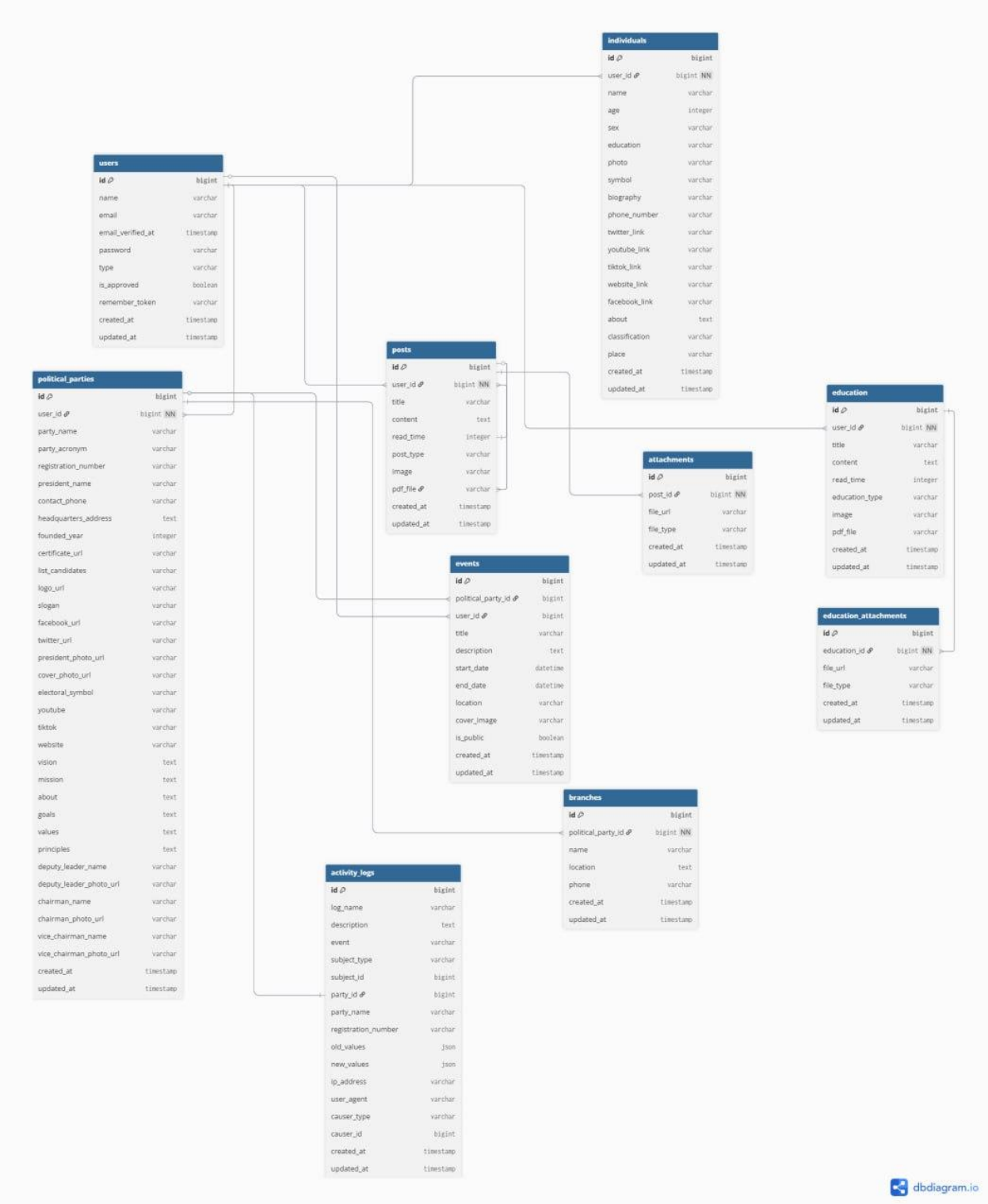
4.7.4 Component Diagram



4.7.5 Deployment Diagram



4.7.6 Database Design



5 Conclusion and recommendation

5.1 Conclusion

In conclusion, the development and successful deployment of the Mirchaye Election Information Platform represent a significant milestone in leveraging technology to strengthen democratic processes in Ethiopia. This project was not merely a software development endeavor but a comprehensive civic-tech initiative designed to combat misinformation, enhance transparency, and foster inclusive voter engagement. The platform stands as a testament to the power of user-centric design, robust technical architecture, and a unwavering commitment to political neutrality and data privacy.

The meticulous planning, from the initial Software Requirements Specification (SRS) and Business Model Canvas to the iterative development sprints, provided a clear and effective roadmap. This structured approach ensured the system was built to the precise specifications of its diverse stakeholders: the National Election Board of Ethiopia (NEBE), political parties, and, most importantly, the Ethiopian citizenry. The successful implementation of a multilingual, no-login-required system directly addresses critical barriers to information access, ensuring equitable participation for voters across all geographical and linguistic backgrounds.

Furthermore, the technical execution—building a secure Laravel backend, a responsive Next.js admin dashboard, and a cross-platform Flutter mobile application—has resulted in a scalable, high-performance ecosystem. The strategic focus on caching, security hardening, and offline functionality ensures the platform's reliability and resilience during the high-demand periods of an electoral cycle. This project has provided an invaluable real-world learning experience, transforming theoretical knowledge into practical, impactful application.

Ultimately, the Mirchaye platform is more than a technological tool; it is a vital civic resource. It embodies a proactive innovation in electoral communication and stands as a powerful contribution to fostering a more informed, engaged, and empowered electorate in Ethiopia, thereby strengthening the very foundations of the nation's democracy.

5.2 Recommendation for Future Internship Program

- Offer Continue Project-Based Learning**:** The model of embedding interns into a single, significant project like Mirchaye is immensely valuable. Future programs should continue this practice, as it provides depth of experience, a sense of ownership, and the satisfaction of contributing to a completed, live product.

- Formalize Agile/Scrum Training: Introduce a dedicated workshop on Agile methodologies and Scrum ceremonies at the internship's outset. This would help interns more quickly understand sprint planning, daily stand-ups, and retrospectives, maximizing their effectiveness within the team structure.

- Incorporate Frontend-Backend Integration Sessions: Schedule structured pairing sessions between interns working on different stack components (e.g., backend API developers with frontend/mobile developers). This would enhance understanding of full-stack integration and improve the quality of API design and consumption.

5.2.1 Recommendation for Future Internship Program

- Offer Structured Training: Future internship programs should include structured training sessions or workshops at the beginning to help interns acquire essential skills and familiarize themselves with the company's processes and tools.

-Mentorship: Assign dedicated mentors to interns, providing guidance and support throughout the internship. A mentorship program can enhance the learning experience and help interns acclimate to the company culture.

-Real-World Projects: Incorporate more real-world projects into the internship, allowing interns to work on tasks that contribute directly to the company's goals. This hands-on experience is invaluable for skill development.

5.2.2 Recommendation for Hosting Company

- Expand Open-Source Contribution: Consider open-sourcing non-critical, reusable components developed during projects like Mirchaye (e.g., the multilingual Laravel module, specific Flutter widgets). This would bolster Ewenet's reputation as a thought leader in Ethiopia's tech community and attract talent.

- Implement a "Deployment Playbook": Document the end-to-end deployment process (from local development to cPanel/staging and production) encountered in this project into a standardized playbook. This will streamline future project go-lives and serve as a training document for new hires and interns.

- Foster Client-Facing Exposure**:** Where appropriate, provide interns with opportunities to sit in on client meetings (e.g., with NEBE officials). Understanding client needs and constraints firsthand is an invaluable professional development experience.

5.2.3 Recommendation for Department

- Clearly Defined Roles: Ensure that interns have well-defined roles and responsibilities, with specific objectives and project goals.

- Provide Challenging Projects: Assign interns to challenging projects that allow them to apply their knowledge and grow their skills.

- Regular Check-Ins: Schedule regular check-in meetings with interns to provide guidance, discuss progress, and address any concerns.

- Collaborative Environment: Foster a collaborative and supportive team environment where interns feel comfortable asking questions and seeking assistance.

- Knowledge Sharing: Encourage knowledge sharing among interns and team members to enhance the collective learning experience.

5.2.4 Recommendation for University:

- Curriculum Alignment: Collaborate with the hosting company to align the internship program with the university's curriculum, ensuring that students can apply what they've learned in the classroom to real-world projects.

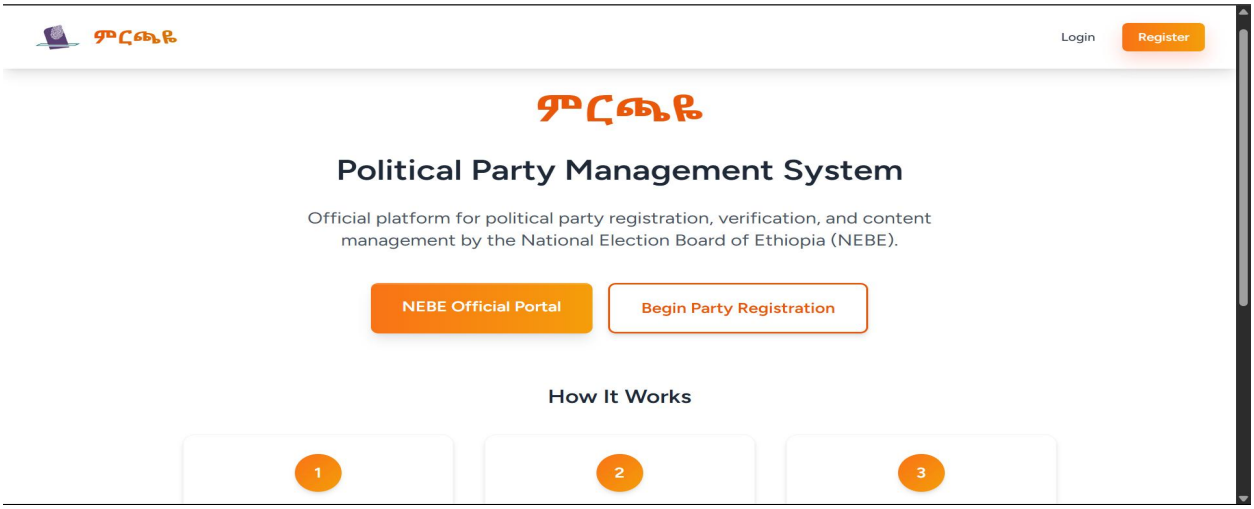
- Continuous Feedback: Establish a feedback loop between the university and the hosting company to exchange insights and assess the performance of interns.

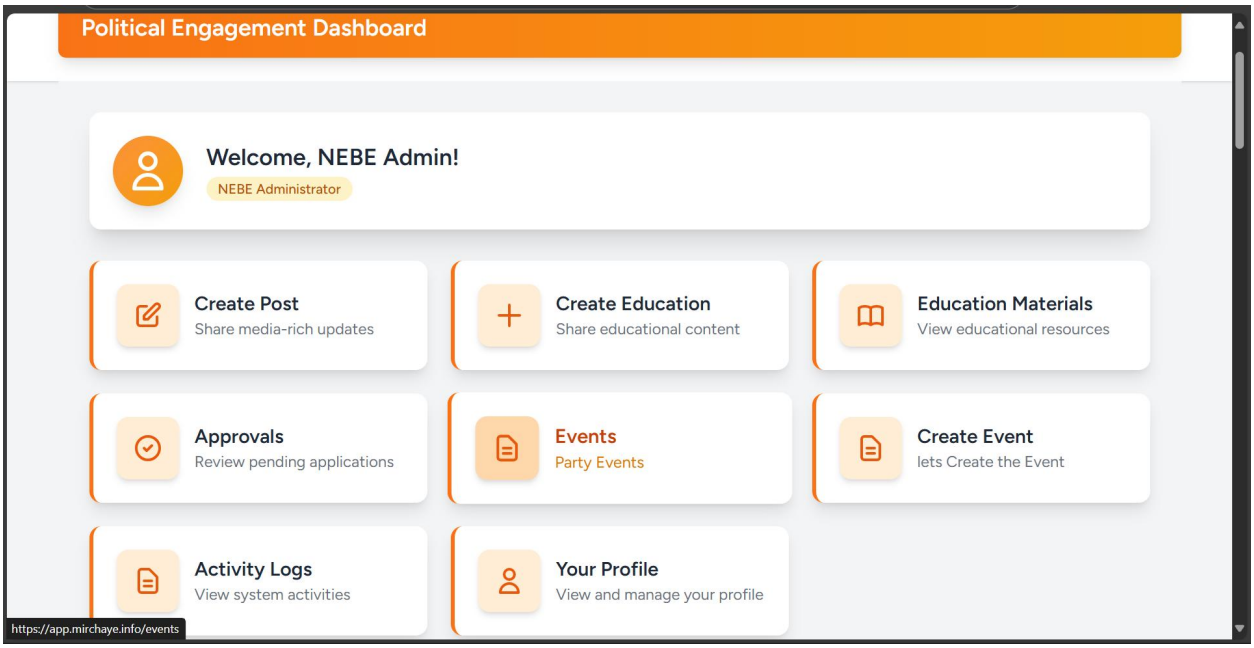
- Skill Development Focus: Emphasize the importance of skill development and practical experience, encouraging students to seek internships that align with their career goals.

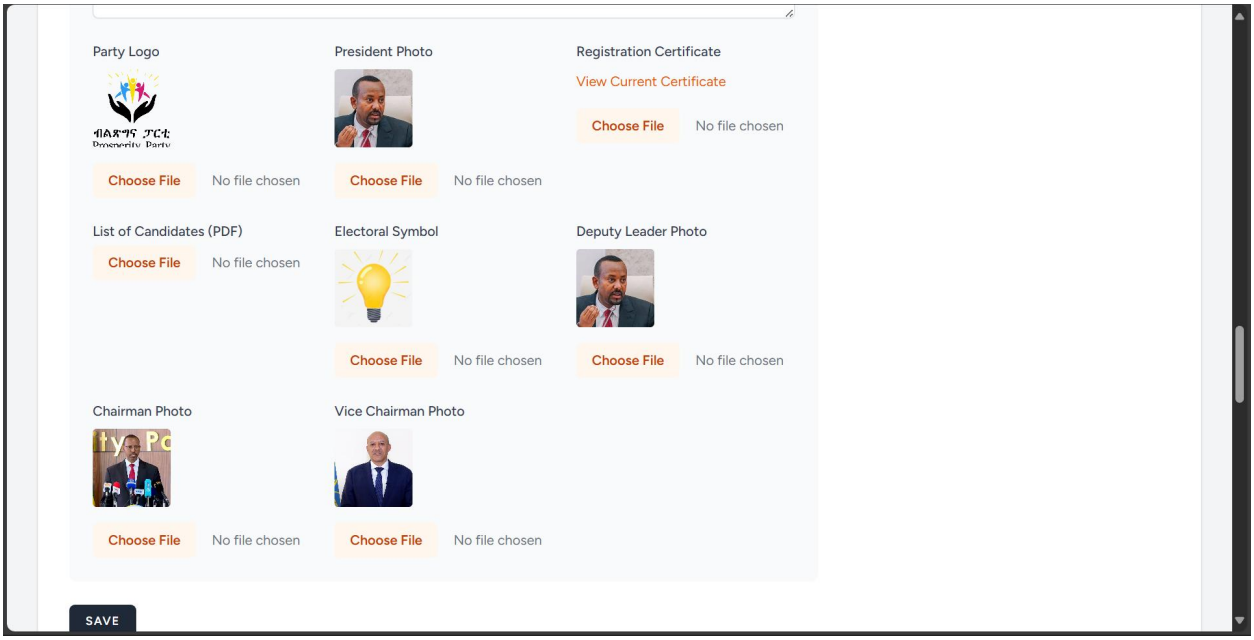
- Career Counseling: Offer career counseling and guidance to help students make informed decisions about their internship choices.

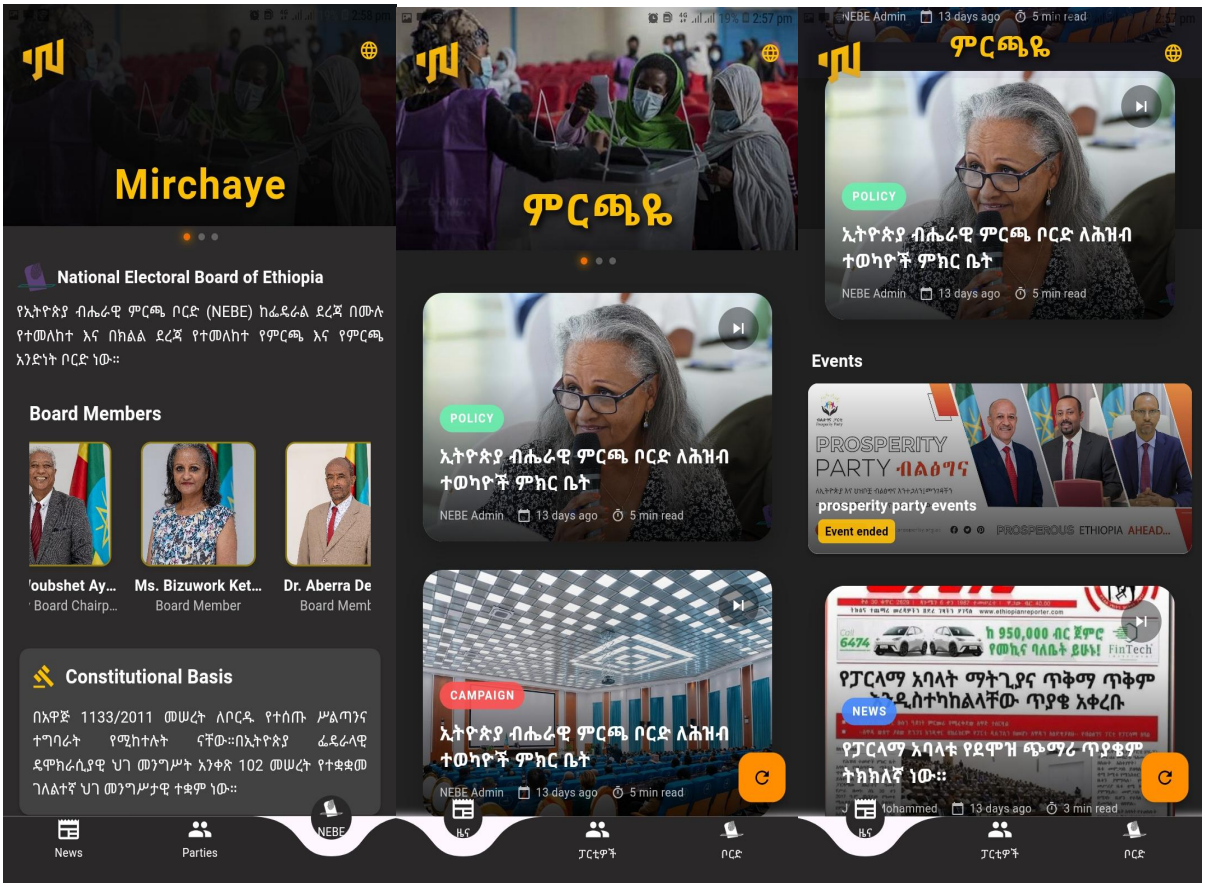
- Encourage Soft Skills: Promote the development of soft skills such as communication, teamwork, and problem-solving, which are essential for success in any internship or job.

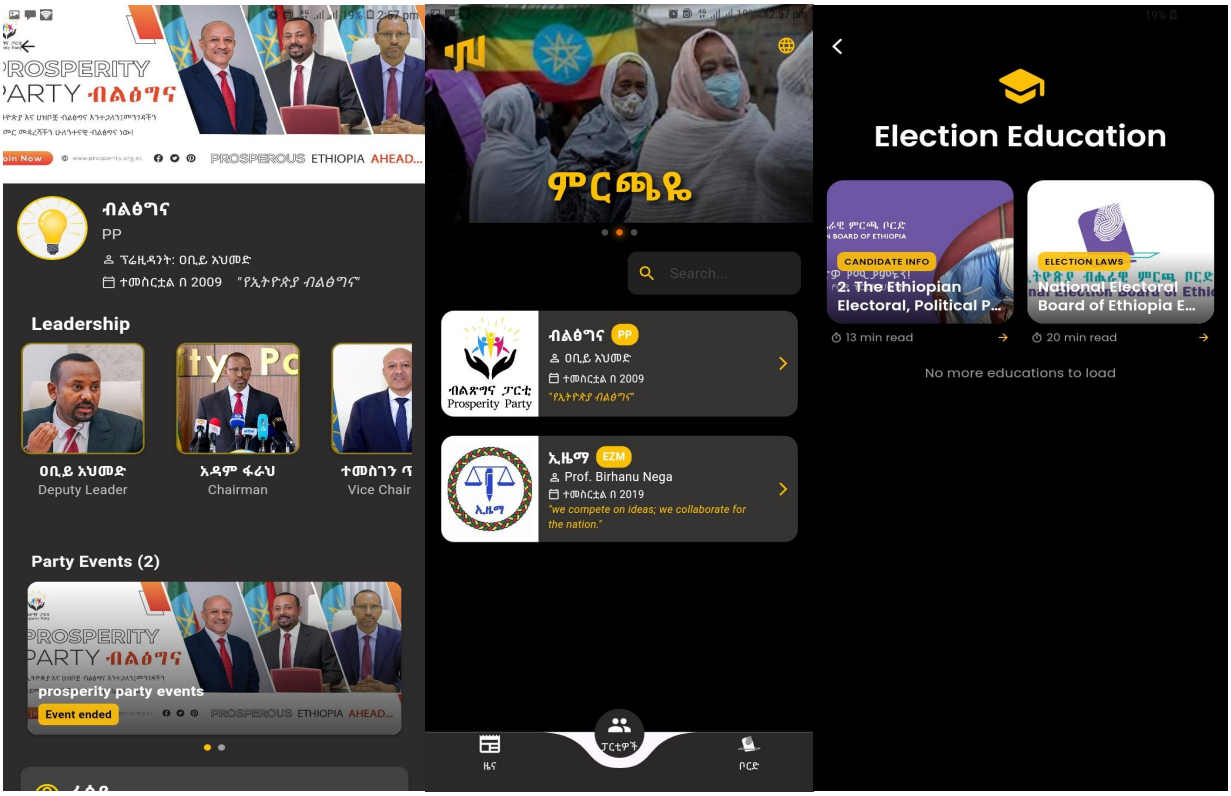
6 Appendix


ምርጫዬ
Login
Register
ምርጫዬ
Political Party Management System
Official platform for political party registration, verification, and content management by the National Election Board of Ethiopia (NEBE).
NEBE Official Portal
Begin Party Registration
How It Works
1
2
3

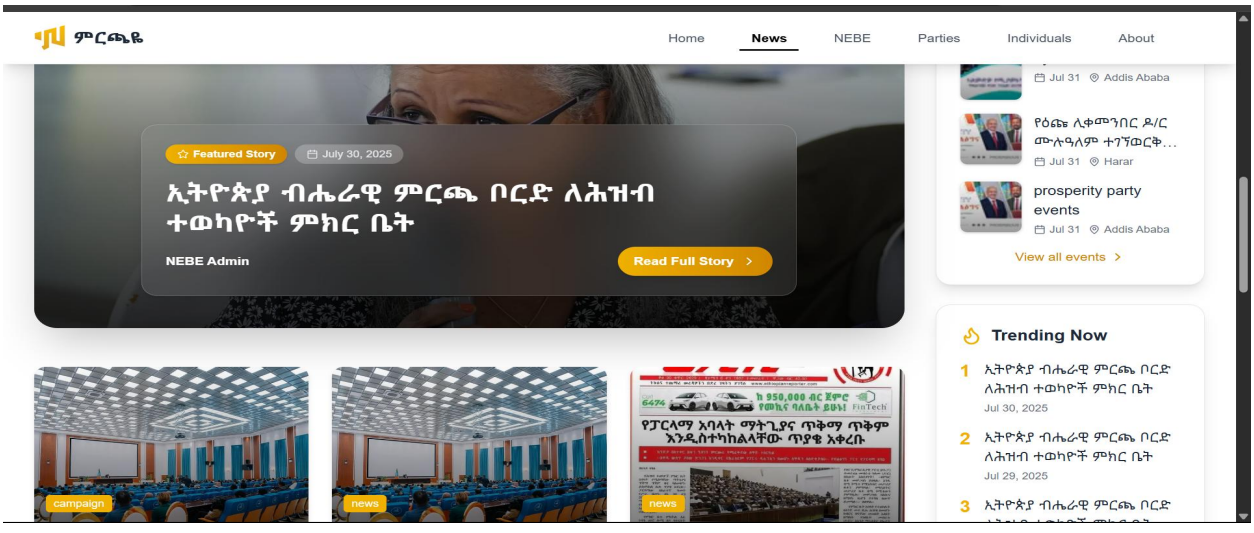

Political Engagement Dashboard
Welcome, NEBE Admin!
NEBE Administrator
Create Post
Share media-rich updates
Create Education
Share educational content
Education Materials
View educational resources
Approvals
Review pending applications
Events
Party Events
Create Event
lets Create the Event
Activity Logs
View system activities
Your Profile
View and manage your profile
https://app.mirchaye.info/events


Party Logo
ብልጽግና ፓርቲ
Prosperity Party
Choose File
No file chosen
President Photo
Choose File
No file chosen
Registration Certificate
View Current Certificate
Choose File
No file chosen
List of Candidates (PDF)
Choose File
No file chosen
Electoral Symbol
Choose File
No file chosen
Deputy Leader Photo
Choose File
No file chosen
Chairman Photo
Choose File
No file chosen
Vice Chairman Photo
Choose File
No file chosen
SAVE


Mirchaye
National Electoral Board of Ethiopia
የኢትዮጵያ ብሔራዊ ምርጫ ቦርድ (NEBE) ከፌዴራል ደረጃ በሙሉ የተመለከተ እና በክልል ደረጃ የተመለከተ የምርጫ እና የምርጫ አንድነት ቦርድ ነው።
Board Members
'oubshet Ay...
Board Chairp...
Ms. Bizuwork Ket...
Board Member
Dr. Aberra De
Board Memt
Constitutional Basis
በአዋጅ 1133/2011 መሠረት ለቦርዱ የተሰጡ ሥልጣንና ተግባራት የሚከተሉት ናቸው።በኢትዮጵያ ፌዴራላዊ ዴሞክራሲያዊ ህገ መንግሥት አንቀጽ 102 መሠረት የተቋቋመ ገለልተኛ ህገ መንግሥታዊ ተቋም ነው።
News
Parties
NEBE
ምርጫዬ
POLICY
ኢትዮጵያ ብሔራዊ ምርጫ ቦርድ ለሕዝብ ተወካዮች ምክር ቤት
NEBE Admin 13 days ago 5 min read
Events
PROSPERITY PARTY ብልፅግና
prosperity party events
Event ended
CAMPAIGN
NEWS
ዜና
ፓርቲዎች
ቦርድ


ብልፅግና
PP
ፕሬዚዳንት: ዐቢይ አህመድ
ተመሰርቷል በ 2009 "የኢትዮጵያ ብልፅግና"
Leadership
ዐቢይ አህመድ
Deputy Leader
አዳም ፋራህ
Chairman
Party Events (2)
prosperity party events
Event ended
PROSPEROUS ETHIOPIA AHEAD...
ምርጫዬ
Search...
ብልፅግና PP
ዐቢይ አህመድ
ተመሰርቷል በ 2009
"የኢትዮጵያ ብልፅግና"
ኢዜማ EZM
Prof. Birhanu Nega
ተመሰርቷል በ 2019
"we compete on ideas; we collaborate for the nation."
ዜና
ፓርቲዎች
ቦርድ
Election Education
CANDIDATE INFO
2. The Ethiopian Electoral, Political P...
13 min read
ELECTION LAWS
National Electoral Board of Ethiopia E...
20 min read
No more educations to load


ምርጫዬ
Home
News
NEBE
Parties
Individuals
About
ስለ ብሔራዊ ምርጫ ቦርድ


ምርጫዬ
Home
News
NEBE
Parties
Individuals
About
Featured Story
July 30, 2025
ኢትዮጵያ ብሔራዊ ምርጫ ቦርድ ለሕዝብ ተወካዮች ምክር ቤት
NEBE Admin
Read Full Story
Jul 31 Addis Ababa
የዕጩ ሊቀመንበር ዶ/ር ሙሉዓለም ተገኝወርቅ...
Jul 31 Harar
prosperity party events
Jul 31 Addis Ababa
View all events
Trending Now
1 ኢትዮጵያ ብሔራዊ ምርጫ ቦርድ ለሕዝብ ተወካዮች ምክር ቤት
Jul 30, 2025
2 ኢትዮጵያ ብሔራዊ ምርጫ ቦርድ ለሕዝብ ተወካዮች ምክር ቤት
Jul 29, 2025
3 ኢትዮጵያ ብሔራዊ ምርጫ ቦርድ
campaign
news
የፓርላማ አባላት ማትጊያና ጥቅማ ጥቅም እንዲሰተካከልላቸው ጥያቄ አቀረቡ
news

7. Reference

- ✓ **National Election Board of Ethiopia (NEBE).**
  ***Official Website*: https://nebe.org.et**
- ✓ **NEBE Facebook Page (for official press releases and updates).**
  https://www.facebook.com/NationalElectionBoardEthiopia/
- ✓ **Laravel PHP Framework Documentation: *https://laravel.com/docs***
- ✓ **Flutter Framework Documentation: *https://flutter.dev/docs***
- ✓ **Next.js React Framework Documentation: *https://nextjs.org/docs***
- ✓ **RESTful API Design — Best Practices: *https://restfulapi.net***
- ✓ **Ethiopian Election Observation Reports – African Union (AU) & EU Reports:**
  https://au.int/en/documents*:*https://www.eeas.europa.eu/